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GALOIS REPRESENTATIONS AND MODULARITY FOR TWISTED MOMENTS OF KLOOSTERMAN SUMS

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Abstract. Associated with the classical Kloosterman sums, we consider the quadratic twisted symmetric power moments. We investigate the cohomological properties and construct the motives over the field of the rationals underlying such moments. We determine the associated modular forms in the lower degree cases.

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1. Introduction

1.1. **The moments.** Let p be a prime number and $\mathbb{F}_p$ the finite field of p elements. Fix a nontrivial additive character $\psi\colon \mathbb{F}_p \to \mathbb{C}^\times$. Fix an algebraic closure $\overline{\mathbb{F}}_p$ of $\mathbb{F}_p$. For any finite field of q elements $\mathbb{F}_q$ in $\overline{\mathbb{F}}_p$, denote $\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}\colon \mathbb{F}_q \to \mathbb{F}_p$ the trace map. For any $a \in \mathbb{F}_q^\times$, the *Kloosterman sum* $\mathrm{Kl}_2(q;a)$ is the exponential sum given by

$$\mathrm{Kl}_2(q;a) = \sum_{x\in\mathbb{F}_q^\times} \psi(\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}(x + ax^{-1})). \tag{1}$$

It is known that, for $a \in \mathbb{F}_q^\times$, there exist algebraic integers α_a, β_a with $\alpha_a\beta_a = q$ such that

$$\mathrm{Kl}_2(q^n;a) = -(\alpha_a^n + \beta_a^n), \quad n \in \mathbb{Z}_{>0}.$$

For a positive integer k, the *symmetric power of degree k*, $\mathrm{Kl}_2^{\mathrm{Sym}^k}(q;a)$, of the Kloosterman sum is defined as

$$\mathrm{Kl}_2^{\mathrm{Sym}^k}(q;a)=\sum_{i=0}^{k}\alpha_a^{k-i}\beta_a^i.$$

(Notice however the sign change $\mathrm{Kl}_2^{\mathrm{Sym}^1}(q;a)=-\mathrm{Kl}_2(q;a)$ in the $k=1$ case.) Let $\kappa_2\colon \mathbb{F}_q^\times\to\mathbb{C}^\times$ be the nontrivial quadratic character. The *symmetric power moment*, $\mathsf{m}_k^+(q)$, and the (quadratic) *twisted symmetric power moment of degree k*, $\mathsf{m}_k(q)$, of the Kloosterman sums are defined respectively as

$$\mathsf{m}_k^+(q)=\sum_{a\in\mathbb{F}_q^\times}\mathrm{Kl}_2^{\mathrm{Sym}^k}(q;a),\quad \mathsf{m}_k(q)=\sum_{a\in\mathbb{F}_q^\times}\kappa_2(a)\,\mathrm{Kl}_2^{\mathrm{Sym}^k}(q;a).$$

Our main focus in this paper is on the twisted moments, so we choose to lighten the notation for the twisted case. It is easy to check that these moments $\mathsf{m}_k^+(q),\mathsf{m}_k(q)$ are indeed integers.

We also consider a related exponential sum

$$\widetilde{\mathrm{Kl}}_2(q;a)=\sum_{y\in\mathbb{F}_q^\times}\psi(\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}\,a(y+y^{-1})),\quad a\in\mathbb{F}_q^\times,$$

which has a similar decomposition

$$\widetilde{\mathrm{Kl}}_2(q^n;a)=-(\tilde\alpha_a^n+\tilde\beta_a^n),\quad n\in\mathbb{Z}_{>0},$$

with $\tilde\alpha_a\tilde\beta_a=q$. (In fact, one has $\widetilde{\mathrm{Kl}}_2(q;a)=\sum_{x\in\mathbb{F}_q^\times}\psi(\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}(x+a^2x^{-1}))=\mathrm{Kl}_2(q;a^2)$ via the substitution $y=a^{-1}x\in\mathbb{F}_q^\times$.) Define accordingly the associated symmetric power moment

$$\widetilde{\mathsf{m}}_k(q)=\sum_{a\in\mathbb{F}_q^\times}\sum_{i=0}^{k}\tilde\alpha_a^{k-i}\tilde\beta_a^i.$$

Since

$$\mathsf{m}_k^+(q)+\mathsf{m}_k(q)=2\sum_{b\in(\mathbb{F}_q^\times)^2}\mathrm{Kl}_2^{\mathrm{Sym}^k}(q;b)$$

and

$$2\sum_{b\in(\mathbb{F}_q^\times)^2}\psi(\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}(x+bx^{-1}))^k=\sum_{a\in\mathbb{F}_q^\times}\psi(\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}\,a(y+y^{-1}))^k$$

by writing $b=(\pm a)^2, x=ay$, we then obtain

$$\widetilde{\mathsf{m}}_k(q)=\mathsf{m}_k^+(q)+\mathsf{m}_k(q)\tag{2}$$

after turning the symmetric power $\sum u^{k-i}v^i$ into a combination of powers $(u+v)^j$ for $0\le j\le k$ with $uv=q$. For instance, by direct computation, one has

$$\mathsf{m}_1^+(q)=-1,\quad \widetilde{\mathsf{m}}_1(q)=-1-\kappa_2(-1)q,\quad \mathsf{m}_2^+(q)=-1,\quad \widetilde{\mathsf{m}}_2(q)=-1-q.$$

Thus,

$$\mathsf{m}_1(q)=-\kappa_2(-1)q,\quad \mathsf{m}_2(q)=-q.\tag{3}$$

We also have

$$\mathsf{m}_3^+(q)=-1-\kappa_2(-3)q^2,\quad \mathsf{m}_4^+(q)=-1-q^2,$$

for $p \geq 5$ and $p \geq 3$, respectively.

Evans's modularity conjectures. These are the explicit descriptions in the next four cases in k for $\mathsf{m}_k^+(p)$ and $\mathsf{m}_k(p)$. Based on extensive computations, Ron Evans observes that the sequences of (the negative of the pure parts of) the moments $\{\mathsf{m}_k^+(p)\}_p$ for $k = 5, 6, 8$ and $\{\mathsf{m}_k(p)\}_p$ for $k = 3, 4, 6$ coincide, up to constant powers of p, with the Fourier coefficients of certain cuspidal Hecke eigenforms of weights $k-2$ and k, respectively, at least for $p > k$. In addition, the moments $\{\mathsf{m}_7^+(p)\}_{p>7}$ and $\{\mathsf{m}_5(p)\}_{p>5}$ are obtained from the symmetric squares of Hecke eigenforms of weight 3, twisted by constant powers of p and Dirichlet characters. See [Eva10, §4] where our moments $\mathsf{m}_k^+(p)$ and $\mathsf{m}_k(p)$ are denoted by $T_k(1,0)$ and $T_k(\phi,0) = T_k(\phi,0,1)$, respectively. See Theorem 1.2.2 and also Remarks 3.2.1 and 3.2.6 below for the precise statements in the case $\mathsf{m}_k(p)$. In the appearance of his paper, this modularity for $\mathsf{m}_5^+, \mathsf{m}_6^+, \mathsf{m}_3, \mathsf{m}_4$ has been obtained by various authors as mentioned therein (see also the note in [Eva10, p.530] for m_4).

Galois representations. Using ideas from the geometric Langlands program and tools including affine Grassmannians and homogeneous Fourier transforms, Yun [Yun15] constructs a continuous ℓ-adic representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, for each prime ℓ, which is a subquotient of the étale cohomology group of a certain projective smooth variety over $\mathbb{Q}$, whose trace of a geometric Frobenius element at an unramified prime p coincides with (the pure part of) the negative moment $-\mathsf{m}_k^+(p)$ for each fixed k. Applying Serre's modularity conjecture [Ser87], proved by Khare and Wintenberger [KW09a, KW09b], to such Galois representations, Yun then proceeds to reprove the modularity for the moments $\mathsf{m}_5^+(p)$ and $\mathsf{m}_6^+(p)$, confirm Evans's modularity conjecture for $\mathsf{m}_7^+(p)$, and reduce that for $\mathsf{m}_8^+(p)$ to a finite calculation, which is done by Vincent in an appendix of the same paper.

On the other hand, aiming to understand the shape of the completed L-function attached to the pure part of the moments $\mathsf{m}_k^+(p)$ for each k and its functional equation, observed by Broadhurst and Roberts [Bro17] via intensive numerics, Fresán, Sabbah and Yu [FSY22] develop a different approach for the arithmetic structures of $\mathsf{m}_k^+(p)$ originated from a viewpoint of (irregular) Hodge theory. The authors obtain an explicit hypersurface $\mathscr{K}$ in a torus whose étale cohomology in the middle degree contains the Galois representation constructed by Yun, up to semisimplification, as a direct summand. The geometric model is simple enough to allow the authors to extract detailed information on ramification behaviors at bad primes for $\mathsf{m}_k^+(p)$, at least for $p > 2$. Moreover, by an investigation of the characteristic zero analogues of the Kloosterman sheaf and its symmetric powers, the Hodge numbers underlying (the motive of) the Galois representation are determined. With the current advances of automorphic theory, this is sufficient to conclude that the Galois representations are potentially automorphic.

In addition, both approaches [Yun15] and [FSY22] apply to the symmetric power moments $\mathsf{m}_k^{(n+1),+}(p)$ of Kloosterman sums of higher rank Kl_{n+1} so one obtains the associated Galois representations. (In fact, Yun constructs such Galois representations for V-moments of the generalized Kloosterman sheaf $\mathrm{Kl}_{\widehat{G}}$ for irreducible representations V of an almost simple split algebraic group $\widehat{G}$ over $\mathbb{Q}_\ell$. The moments $\mathsf{m}_k^{(n+1),+}(p)$ correspond to $\widehat{G} = \mathrm{SL}_{n+1}$ with V the k-th symmetric power of the standard representation.) Qin [Qin24] then uses the irregular Hodge approach as in [FSY22] to investigate the Hodge structures underlying $\mathsf{m}_k^{(n+1),+}(p)$, and obtains the potential automorphy in certain cases. Adapting the modularity methods

similar to [Yun15], Qin proves the Evans type formulas relating the moments $\mathsf{m}_k^{(n+1),+}(p)$ and explicit modular forms in the rank two cases, including all the examples listed in [Yun15, §1.3] for the group $\widehat{G} = \mathrm{SL}_{n+1}$.

We remark that an explicit *virtual* scheme over $\mathbb{Z}$ whose reduction modulo p determines $\mathsf{m}_k^{(n+1),+}(p)$ is constructed in [FW08].

1.2. **Main results.** The main goal of this paper is to investigate the arithmetic of the sequence of twisted moments $\{\mathsf{m}_k(p)\}$ as p varies. In particular, we extend the Galois representation theoretic parts and the modularity results in [Yun15] and [FSY22] to the twisted setting. We have the following two theorems.

Theorem 1.2.1. *For each $k \geq 1$, let $d = \lfloor (k+1)/2 \rfloor - \delta_{2+4\mathbb{Z}}(k)$ where $\delta_{2+4\mathbb{Z}}$ is the characteristic function supported on $2 + 4\mathbb{Z} \subset \mathbb{Z}$. There exists a compatible family of continuous ℓ-adic representations for prime numbers ℓ*

$$\rho_{k,\ell} \colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_d(\mathbb{Q}_\ell),$$

which is unramified and satisfying

$$\mathrm{tr}\left(\rho_{k,\ell}(\mathrm{Frob}_p)\right) = -\mathsf{m}_k(p) - \delta_{2+4\mathbb{Z}}(k) p^{k/2}$$

at prime $p \neq \ell$ and $p > k$ (resp., $p > k/2$) if k is odd (resp., k is even). Here Frob_p is a geometric Frobenius element at p in $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Moreover, $\rho_{k,\ell}$ is self-dual, up to a Tate twist, is de Rham above ℓ and crystalline if $\ell > \max\{k, 2\}$ (resp., $\ell > k/2$) when k is odd (resp., k is even), and is potentially automorphic.

In the following, the `LMFDB` is referred to the online database [LMF21] of L-functions and modular forms.

Theorem 1.2.2. *For a cuspidal Hecke eigenform f on some congruence subgroup, let $f(q) = \sum_{n=1}^{\infty} a_f(n) q^n$ be its q-expansion. The following modularity holds.*

(i) *Let f be the modular form $f_{48.3.e.a}$ (CM by $\mathbb{Q}(\sqrt{-3})$) in `LMFDB`. Then, for $p \geq 5$,*

$$\mathsf{m}_3(p) = -a_f(p) \cdot p. \tag{4}$$

(ii) *Let f be the modular form $f_{8.4.a.a}$ in `LMFDB`. Then, for each odd prime p,*

$$\mathsf{m}_4(p) = -a_f(p) \cdot p$$

(iii) *Let f be the modular form $f_{60.3.b.a}$ in `LMFDB` (with $a_f(n) \in \mathbb{Q}(\sqrt{5}, i)$). Then, when $p \geq 7$,*

$$\mathsf{m}_5(p) = -\left(\frac{-1}{p}\right) a_f(p)^2 \cdot p + \left(\frac{15}{p}\right) p^3, \tag{5}$$

where $\left(\frac{\cdot}{p}\right)$ is the Legendre symbol.

(iv) *Let f be the modular form $f_{24.6.a.a}$ in `LMFDB`. Then, when $p \geq 5$,*

$$\mathsf{m}_6(p) = -a_f(p) \cdot p - p^3.$$

This confirms Evans's modularity conjectures for $\mathsf{m}_k(p)$.

Theorem 1.2.1 is a simplified version of the contents of Propositions 3.1.2, 3.1.3 and 3.1.4, and Corollary 3.1.6 below. In fact, such Galois representations are already constructed in [FSY22], at least implicitly, as a direct summand of the étale cohomology of the same

hypersurface $\mathscr{K}$ for $\mathsf{m}_k^+(p)$ mentioned previously. Essentially, the only missing point is to pin down the determinants of the representations $\rho_{k,\ell}$. The latter relies on the results obtained in §2, in which we investigate in detail the étale cohomology associated with $\mathsf{m}_k(p)$ for fixed k and p. The consideration extends the work [FW10] for the untwisted case $\mathsf{m}_k^+(p)$. The modularity Theorem 1.2.2 is obtained using similar arguments as in [Yun15].

The rest of the paper is organized as follows. In §2, we recall the construction of the Kloosterman sheaf Kl_2 on $\mathbb{G}_{\mathrm{m}}$ in positive characteristic. We investigate in detail the local structures of the sheaf $\widetilde{\mathrm{Kl}}_2$, which is the pullback of Kl_2 by a double cover of $\mathbb{G}_{\mathrm{m}}$, and use them to obtain the cohomological properties for $\mathsf{m}_k(p)$ for fixed positive k and odd p. The approach and the main results in this section are based on the work [FW10] in an essential way. In §3, we construct the associated Galois representation for $\mathsf{m}_k(p)$ for fixed k, mainly by extracting the necessary information from [FSY22], and draw some arithmetic and analytic consequences from the potential automorphy of the representation. We then prove Theorem 1.2.2 relating $\mathsf{m}_k(p)$ for $k = 3, 4, 5, 6$ to modular forms using Serre's modularity conjecture.

In Appendix §A.1, we indicate that one can obtain the main results in §2 by direct inspections of the quadratic twists of the symmetric powers of Kl_2, instead of going up to $\widetilde{\mathrm{Kl}}_2$. Still, this approach relies on the calculations in [FW10]. In §A.2, we record the relationship between the twisted Kloosterman sums and hypergeometric type character sums and their modularity. We record some facts about the associated L-functions in §A.3.

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2. Cohomology for the moments

Fix an odd prime p and a positive integer k. We investigate in detail the twisted moments $\mathsf{m}_k(p)$ cohomologically in this section.

2.1. **Étale sheaves.** We work over the base field $\mathbb{F}_p$. When we need to specify a variable for the torus over $\mathbb{F}_p$, we set $\mathbb{G}_{\mathrm{m},x} = \operatorname{Spec} \mathbb{F}_p[x, x^{-1}]$, indicating the variable x in the subscript; otherwise we omit it. A similar convention applies to the affine line $\mathbb{A}^1$ and the projective line $\mathbb{P}^1$. We recall briefly the construction of the Kloosterman sheaf and some basic properties that we take for granted.

We fix a rational prime ℓ, distinct from p, and a nontrivial additive character

$$\psi \colon \mathbb{F}_p \longrightarrow \overline{\mathbb{Q}}_\ell^\times.$$

(One can replace the coefficient field $\overline{\mathbb{Q}}_\ell$ by the extension of $\mathbb{Q}_\ell$ joined with p-th roots of 1.) We extend ψ to a character of any finite field extension of $\mathbb{F}_p$ by composing with the trace map. The Artin–Schreier cover $\mathbb{A}^1_t \to \mathbb{A}^1_s$ induced from $s \mapsto t^p - t$ is a Galois cover; it then corresponds to a quotient $\pi_1^{\text{ét}}(\mathbb{A}^1_s) \to \mathrm{Aut}(\mathbb{A}^1_t/\mathbb{A}^1_s) = \mathbb{F}_p$ of the étale fundamental group of $\mathbb{A}^1$. Composing with ψ^{-1} to obtain a character of $\pi_1^{\text{ét}}(\mathbb{A}^1_s)$, we get the *Artin–Schreier sheaf* $\mathscr{L}_\psi$, which is a lisse sheaf on $\mathbb{A}^1$ with Swan conductor $\mathrm{sw}_\infty(\mathscr{L}_\psi) = 1$ at ∞. It is also characterized by the following trace function. For a closed point $x \in \mathbb{A}^1(\mathbb{F}_q) = \mathbb{F}_q$, let Frob_x be a geometric Frobenius at x. Then $\mathrm{Frob}_x|_{\mathscr{L}_\psi} = \psi(x)$. For a regular function $f : X \to \mathbb{A}^1$ of X over $\mathbb{F}_p$,

let $\mathscr{L}_\psi(f)$ be the pull-back $f^*\mathscr{L}_\psi$. For a geometric Frobenius Frob_x at $x \in X(\mathbb{F}_q)$, we then have $\mathrm{Frob}_x|_{\mathscr{L}_\psi(f)} = \psi(f(x))$. One has the duality

$$\mathscr{L}_\psi(f)^\vee = \mathscr{L}_\psi(-f). \tag{6}$$

Consider the diagram

$$\mathbb{G}_{\mathrm{m},z} \xleftarrow{\pi} \mathbb{G}_{\mathrm{m},z} \times \mathbb{G}_{\mathrm{m},x} \xrightarrow{f} \mathbb{A}^1_s$$

where π is the projection to the first component and f is induced by $s \mapsto x + zx^{-1}$. The *Kloosterman sheaf* is the lisse sheaf Kl_2 of rank two on $\mathbb{G}_\mathrm{m}$ over $\mathbb{F}_p$ defined as

$$\mathrm{Kl}_2 = \mathbf{R}^1\pi_!\,\mathscr{L}_\psi(f) = \mathbf{R}^1\pi_*\,\mathscr{L}_\psi(f).$$

It is pure of weight one. For a closed point $a \in \mathbb{G}_\mathrm{m}(\mathbb{F}_q) = \mathbb{F}_q^\times$, one has $\mathrm{tr}(\mathrm{Frob}_a \mid \mathrm{Kl}_2) = -\mathrm{Kl}_2(q;a)$, the negative of the Kloosterman sum valued in $\overline{\mathbb{Q}}_\ell$, instead of $\mathbb{C}$, as defined in (1). Since the automorphism $(z,x) \mapsto (-z,-x)$ on $\mathbb{G}_{\mathrm{m},z} \times \mathbb{G}_{\mathrm{m},x}$ sends f to $-f$, the duality (6) induces a perfect alternating pairing

$$\mathrm{Kl}_2 \otimes \mathrm{Kl}_2 \longrightarrow \mathbf{R}^2\pi_!\overline{\mathbb{Q}}_\ell = \overline{\mathbb{Q}}_\ell(-1). \tag{7}$$

On the other hand, let

$$[2]\colon \mathbb{G}_{\mathrm{m},t} \longrightarrow \mathbb{G}_{\mathrm{m},z}$$

be the double cover induced by $z \mapsto t^2$. It yields a surjection $\pi_1^{\text{ét}}(\mathbb{G}_{\mathrm{m},z}) \to \mathrm{Aut}(\mathbb{G}_{\mathrm{m},t}/\mathbb{G}_{\mathrm{m},z}) = \{\pm 1\}$. Composing with the nontrivial character $\{\pm 1\} \subset \overline{\mathbb{Q}}_\ell^\times$, we obtain a lisse sheaf κ_2 on $\mathbb{G}_\mathrm{m}$. One has

$$[2]_*\overline{\mathbb{Q}}_\ell = \overline{\mathbb{Q}}_\ell \oplus \kappa_2.$$

Let $\widetilde{\mathrm{Kl}_2} = [2]^*\,\mathrm{Kl}_2$ and

$$\widetilde{\mathscr{V}_k} = \mathrm{Sym}^k\,\widetilde{\mathrm{Kl}_2}, \quad \mathscr{V}_k^+ = \mathrm{Sym}^k\,\mathrm{Kl}_2, \quad \mathscr{V}_k = \kappa_2 \otimes \mathrm{Sym}^k\,\mathrm{Kl}_2,$$

the k-th symmstric power of $\widetilde{\mathrm{Kl}_2}$ and Kl_2 and its quadratic twist, respectively. So for $a \in \mathbb{G}_\mathrm{m}(\mathbb{F}_q) = \mathbb{F}_q^\times$, one has $\mathrm{tr}(\mathrm{Frob}_a \mid \mathscr{V}_k^+) = \mathrm{Kl}_2^{\mathrm{Sym}^k}(q;a)$, the symmetric power of $\mathrm{Kl}_2(q;a)$, and similarly for $\widetilde{\mathscr{V}_k}$ and $\mathscr{V}_k$. There are $(-1)^k$-symmetric perfect pairings

$$\mathscr{F} \otimes \mathscr{F} \longrightarrow \overline{\mathbb{Q}}_\ell(-k), \quad \mathscr{F} = \widetilde{\mathscr{V}_k}, \mathscr{V}_k^+, \mathscr{V}_k \tag{8}$$

inherited from (7) and the self-duality of κ_2. We have

$$[2]^*\mathscr{V}_k^+ = \widetilde{\mathscr{V}_k}, \quad [2]_*\widetilde{\mathscr{V}_k} = \mathscr{V}_k^+ \oplus \mathscr{V}_k; \tag{9}$$

the letter is the sheaf avatar of (2). By the Lefschetz trace formula, the twisted moment $\mathsf{m}_k(p)$ is determined by the Frobenius action on the étale cohomology with compact support $\mathrm{H}^i_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k)$. For $\mathscr{F} = \widetilde{\mathscr{V}_k}, \mathscr{V}_k^+, \mathscr{V}_k$, there is the duality pairing

$$\mathrm{H}^i_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{F}) \otimes \mathrm{H}^{2-i}_{\text{ét}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{F}) \longrightarrow \mathrm{H}^2_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \overline{\mathbb{Q}}_\ell(-k)) = \overline{\mathbb{Q}}_\ell(-k-1) \tag{10}$$

induced by (8). We also consider the *middle part* of the cohomology

$$\mathrm{H}^i_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{F}) = \mathrm{image}\left\{\mathrm{H}^i_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{F}) \longrightarrow \mathrm{H}^i_{\text{ét}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{F})\right\}$$

where the arrow is the forgetting-support map. It inherits the $(-1)^{k+1}$-symmetric perfect pairing

$$\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{F}) \otimes \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{F}) \longrightarrow \overline{\mathbb{Q}}_\ell(-k-1). \tag{11}$$

In fact, letting $j\colon \mathbb{G}_{\mathrm{m}} \to \mathbb{P}^1$ be the inclusion, one has the identity

$$\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{F}) = \mathrm{H}^1_{\text{ét}}(\mathbb{P}^1_{\overline{\mathbb{F}}_p}, j_*\mathscr{F}), \quad \mathscr{F} = \widetilde{\mathscr{V}}_k, \mathscr{V}_k^+, \mathscr{V}_k. \tag{12}$$

As $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$-modules, these spaces are pure of weight $k+1$.

We record the cohomological properties for $\mathscr{V}_k^+$ obtained in [FW05, FW10] in the following proposition. Let I_0 and I_∞ be the inertia groups of $\pi_1^{\text{ét}}(\mathbb{G}_{\mathrm{m}})$ at the boundary points $0, \infty \in \mathbb{P}^1 \setminus \mathbb{G}_{\mathrm{m}}$, respectively, acting on $\widetilde{\mathscr{V}}_k, \mathscr{V}_k^+, \mathscr{V}_k$. Let $F_p \in \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ be the geometric Frobenius automorphism.

Proposition 2.1.1. *As $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$-modules, there is the exact sequence*

$$0 \longrightarrow (\mathscr{V}_k^+)^{I_0} \oplus (\mathscr{V}_k^+)^{I_\infty} \longrightarrow \mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k^+) \longrightarrow \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k^+) \longrightarrow 0.$$

The space $(\mathscr{V}_k^+)^{I_0}$ is of dimension one, upon which $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ acts trivially. The following properties hold.

(i) *Suppose k is odd. Then $(\mathscr{V}_k^+)^{I_\infty} = 0$ and*

$$\det\left(F_p \mid \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k^+)\right) = p^{\frac{k+1}{2}\delta^+} \left(\frac{-2}{p}\right)^{\lfloor \frac{k}{2p}+\frac{1}{2} \rfloor} \prod_{\substack{0 \le j \le (k-1)/2 \\ p \nmid (2j+1)}} \left(\frac{(-1)^j(2j+1)}{p}\right) \tag{13}$$

where

$$\delta^+ = \dim \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k^+) = \frac{k-1}{2} - \left\lfloor \frac{k}{2p} + \frac{1}{2} \right\rfloor.$$

(ii) *Suppose k is even.*

(a) *$\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ acts on $(\mathscr{V}_k^+)^{I_\infty}$ semisimply and*

$$\det(1 - F_p T \mid (\mathscr{V}_k^+)^{I_\infty}) = (1 - p^{k/2}T)^{\delta_{4\mathbb{Z}}(k) + n_k^+(p)} (1 - (-1)^{(p-1)/2} p^{k/2} T)^{n_k^-(p)}$$

with

$$n_k^+(p) = \left\lfloor \frac{k}{4p} \right\rfloor, \quad n_k^-(p) = \left\lfloor \frac{k}{4p} + \frac{1}{2} \right\rfloor. \tag{14}$$

(b) *We have* $\dim \mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k^+) = \frac{k}{2} - \left\lfloor \frac{k}{2p} \right\rfloor$.

(c) *One has*

$$\det\left(F_p \mid \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k^+)\right) = p^{\frac{k+1}{2}\delta^+}$$

where

$$\delta^+ = \dim \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k^+) = \frac{k}{2} - 2\left\lfloor \frac{k}{2p} \right\rfloor - \delta_{4\mathbb{Z}}(k).$$

The rest of this section is devoted to proving the following theorem. By the decomposition in (9), we have

$$\mathrm{H}^i_{\text{ét},?}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \widetilde{\mathscr{V}}_k) = \mathrm{H}^i_{\text{ét},?}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k^+) \oplus \mathrm{H}^i_{\text{ét},?}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k), \quad ? = \emptyset, \mathrm{c}, \mathrm{mid}. \tag{15}$$

What we do below is indeed to obtain the corresponding results for $\widetilde{\mathscr{V}}_k$ based on the work of Fu and Wan, thanks to the equalities in (9).

Theorem 2.1.2. (i) *Suppose k is odd. We have the canonical isomorphisms*

$$\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\xrightarrow{\sim}\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\xrightarrow{\sim}\mathrm{H}^1_{\text{ét}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k).$$

These spaces are of dimension $\delta=\frac{k+1}{2}-\left\lfloor\frac{k}{2p}+\frac{1}{2}\right\rfloor$. Moreover, one has

$$\det\left(F_p\mid\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\right)=p^{\frac{k+1}{2}\delta}\left(\frac{-1}{p}\right)^{\delta}\left(\frac{2}{p}\right)^{\left\lfloor\frac{k}{2p}+\frac{1}{2}\right\rfloor}\prod_{\substack{0\le j\le(k-1)/2\\ p\nmid(2j+1)}}\left(\frac{(-1)^j(2j+1)}{p}\right).\tag{16}$$

(ii) *Suppose k is even. As $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$-modules, there is the exact sequence*

$$0\longrightarrow\mathscr{V}_k^{I_\infty}\longrightarrow\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\longrightarrow\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\longrightarrow 0.$$

The following properties hold.

(a) *$\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ acts on $\mathscr{V}_k^{I_\infty}$ semisimply and*

$$\det(1-F_pT\mid\mathscr{V}_k^{I_\infty})=(1-p^{k/2}T)^{\delta_{2+4\mathbb{Z}}(k)+n_k^+(p)}(1-(-1)^{(p-1)/2}p^{k/2}T)^{n_k^-(p)}$$

where $n_k^+(p),n_k^-(p)$ are defined in (14).

(b) *We have $\dim\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)=\frac{k}{2}-\left\lfloor\frac{k}{2p}\right\rfloor$.*

(c) *One has*

$$\det\left(F_p\mid\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\right)=p^{\frac{k+1}{2}\delta}$$

where

$$\delta=\dim\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)=\frac{k}{2}-2\left\lfloor\frac{k}{2p}\right\rfloor-\delta_{2+4\mathbb{Z}}(k).$$

Let

$$\begin{aligned}Z_k(p;T)&=\det\left(1-F_pT\mid\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\right),\\ M_k(p;T)&=\det\left(1-F_pT\mid\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\right).\end{aligned}\tag{17}$$

By Lemma 2.3.1 below, one has $\mathsf{m}_k(p)=Z_k'(p;0)$. Due to the self-duality (11), we have the functional equation

$$M_k(p;T)=cT^\delta M_k(p;(p^{k+1}T)^{-1})$$

where $\delta=\deg M_k(p;T)$, and

$$c=\det\left(-F_p\mid\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\right)=(-1)^\delta\det\left(F_p\mid\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathscr{V}_k)\right).\tag{18}$$

2.2. **Local structures.** We investigate the local structures of $\widetilde{\mathscr{V}_k}$ by pulling back the results obtained in [FW10] via (9).

Let $j\colon\mathbb{A}^1\to\mathbb{P}^1$ be the smooth compactification over $\mathbb{F}_p$. For a closed point $x\in|\mathbb{P}^1|$, we let $\mathbb{P}^1_{(x)}$ be the henselization at x, and let η_x be its generic point. Choose a geometric point $\bar\eta_x$ above η_x. Let I_x be the inertia subgroup of the Galois group $\mathrm{Gal}(\bar\eta_x/\eta_x)$.

At the point 0, let $t_\ell\colon I_0\to\mathbb{Q}_\ell(1)=\varprojlim\mu_{\ell^n}(\overline{\mathbb{Q}}_\ell)$ be the ℓ-adic fundamental tame character given by

$$t_\ell(\sigma)=\varprojlim\frac{\sigma(\sqrt[\ell^n]{z})}{\sqrt[\ell^n]{z}}.$$

Here, $\sqrt[\ell^n]{z}$ is any choice of ℓ^n-th root of z on $\bar{\eta}_0$. Fix a lifting $\widetilde{F}_p \in \mathrm{Gal}(\bar{\eta}_0/\eta_0)$ of the geometric Frobenius $F_p \in \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$.

To distinguish the considerations of the sheaves $\mathscr{V}_k^+$ and $\widetilde{\mathscr{V}}_k$, we let $\eta_{\tilde{0}}$ be the corresponding point above η_0 under the double cover $\mathbb{P}^1_t \to \mathbb{P}^1_z$ with a geometric point $\bar{\eta}_{\tilde{0}}$ above $\eta_{\tilde{0}}$. We have the inertia subgroup $I_{\tilde{0}}$ and an element $\widetilde{F}_p$ of $\mathrm{Gal}(\bar{\eta}_{\tilde{0}}/\eta_{\tilde{0}})$ mapping to the geometric Frobenius.

Lemma 2.2.1. *There exists a basis $\{\tilde{f}_i\}_{i=0}^k$ of $\widetilde{\mathscr{V}}_k|_{\eta_{\tilde{0}}}$ such that $\widetilde{F}_p(\tilde{f}_i) = p^i \tilde{f}_i$ and, for $\sigma \in I_{\tilde{0}}$,*

$$\sigma(\tilde{f}_i) = \exp(t_\ell(\sigma)\mathrm{N})\tilde{f}_i, \quad \textit{with } \mathrm{N}(\tilde{f}_i) = \tilde{f}_{i-1}.$$

Here one puts $\tilde{f}_{-1} = 0$. In particular, $\widetilde{\mathscr{V}}_k|_{\eta_{\tilde{0}}}$ is tamely ramified, $\widetilde{\mathscr{V}}_k^{I_{\tilde{0}}} = \langle \tilde{f}_0 \rangle$, and $F_p|_{\widetilde{\mathscr{V}}_k^{I_{\tilde{0}}}} = 1$.

Proof. By [FW10, Lemma 1.2], there exists a basis $\{f_i\}_{i=0}^k$ of the restriction $\mathscr{V}_k^+|_{\eta_0}$ such that $\widetilde{F}_p(f_i) = p^i f_i$ and, for $\sigma \in I_0$,

$$\sigma(f_i) = \exp(t_\ell(\sigma)\mathrm{N})f_i, \quad \text{with } \mathrm{N}(f_i) = f_{i-1}$$

where we set $f_{-1} = 0$. By pulling back to $\eta_{\tilde{0}}$, we obtain the statements. □

We turn to the infinity point. We again distinguish the infinity points ∞ and $\widetilde{\infty}$ for $\mathscr{V}_k^+$ and $\widetilde{\mathscr{V}}_k$, respectively.

Consider the quadratic Gauss sum

$$g = -\sum_{x \in \mathbb{F}_p^\times} \left(\frac{x}{p}\right)\psi(x).$$

It is well-known that

$$g^2 = (-1)^{(p-1)/2}p. \tag{19}$$

We have the natural surjection $\pi_1^{\text{ét}}(\mathbb{G}_\mathrm{m}) \to \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ whose kernel is the geometric fundamental group. Introduce two characters

$$\theta_0, \theta_1 \colon \pi_1^{\text{ét}}(\mathbb{G}_\mathrm{m}) \longrightarrow \overline{\mathbb{Q}}_\ell^\times$$

given by composing characters $\theta_i' \colon \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \to \overline{\mathbb{Q}}_\ell^\times$, respectively, where

$$\theta_0'(F_p) = g, \quad \theta_1'(F_p) = (-1)^{(p-1)/2}.$$

We set

$$\mathcal{K}_i^+ = \mathscr{L}_\psi(-2(k-2i)t), \quad \mathcal{K}_i = \mathscr{L}_\psi(-2(k-2i)t) \otimes \kappa_2, \quad \mathcal{F}_i = \mathcal{F}(\theta_0^k\theta_1^i).$$

Here, for an integer a, $\mathscr{L}_\psi(at)$ is the pullback of $\mathscr{L}_\psi$ on $\mathbb{A}^1_t$ by the map induced by $t \mapsto at$, and $\mathcal{F}(\theta_0^a\theta_1^b)$ is the lisse sheaf corresponding to the character $\theta_0^a\theta_1^b$. Notice that the sheaf $\mathcal{F}_i$ is unramified and κ_2 is tamely ramified at $\mathbb{P}^1 \setminus \mathbb{G}_\mathrm{m}$. For $a \in \mathbb{Z}$, the sheaf $\mathscr{L}_\psi(at)$, lisse on $\mathbb{A}^1$, is ramified at $\widetilde{\infty}$ if and only if $p \nmid a$. In the latter case, it has Swan conductor $\mathrm{sw}_{\widetilde{\infty}}(\mathscr{L}_\psi(at)) = 1$.

Let

$$\Xi_k(p) = \left\{ i \in \mathbb{Z}_{\geq 0} \mid 0 \leq i \leq \lfloor (k-1)/2 \rfloor, k - 2i \equiv 0 \bmod p \right\}. \tag{20}$$

Lemma 2.2.2. *We have the decomposition of $\widetilde{\mathscr{V}}_k|_{\eta_{\widetilde{\infty}}}$ as follows.*

(i) *If $k = 2r+1$ is odd,*

$$\widetilde{\mathscr{V}}_k|_{\eta_\infty} = \bigoplus_{i=0}^{r} \mathcal{V}_i^{\oplus 2}, \quad \mathcal{V}_i = \mathcal{K}_i \otimes \mathcal{F}_{i+1} \tag{21}$$

with Swan conductor $\mathrm{sw}(\mathcal{V}_i) = 1 - \delta_{\Xi_k(p)}(i)$. *In particular,* $\widetilde{\mathscr{V}}_k^{I_\infty} = 0$.

(ii) *In case $k = 2r$ is even,*

$$\widetilde{\mathscr{V}}_k|_{\eta_\infty} = \mathcal{F}_r \oplus \bigoplus_{i=0}^{r-1} (\mathcal{V}_i^+)^{\oplus 2}, \quad \mathcal{V}_i^+ = \mathcal{K}_i^+ \otimes \mathcal{F}_i$$

with Swan conductor $\mathrm{sw}(\mathcal{V}_i^+) = 1 - \delta_{\Xi_k(p)}(i)$. *Moreover,* $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ *acts on* $\widetilde{\mathscr{V}}_k^{I_\infty}$ *semisimply and*

$$\det(1 - F_p T \mid \widetilde{\mathscr{V}}_k^{I_\infty}) = (1 - p^{k/2}T)^{1+2n_k^+(p)}(1 - (-1)^{(p-1)/2}p^{k/2}T)^{2n_k^-(p)} \tag{22}$$

with $n_k^\pm(p)$ defined previously in (14).

Proof. Recall [FW10, Lemma 2.4] that

$$\mathscr{V}_k^+|_{\eta_\infty} = \begin{cases} \displaystyle\bigoplus_{i=0}^{r} [2]_* \mathcal{K}_i \otimes \mathcal{F}_{i+1} & \text{for } k = 2r+1 \text{ odd}, \\ \kappa_2^r \otimes \mathcal{F}_r \oplus \displaystyle\bigoplus_{i=0}^{r-1} [2]_* \mathcal{K}_i^+ \otimes \mathcal{F}_i & \text{for } k = 2r \text{ even}. \end{cases}$$

Pulling back to $\eta_{\widetilde{\infty}}$, the statements follow, except for the last equality (22). For $i \in \Xi_k(p)$, $\mathcal{K}_i^+$ is trivial and $\mathcal{V}_i^+ = \mathcal{F}_i$, so one has

$$\widetilde{\mathscr{V}}_k^{I_\infty} = \mathcal{F}_r \oplus \bigoplus_{i \in \Xi_k(p)} \mathcal{F}_i^{\oplus 2}.$$

By (19), we have $F_p|_{\mathcal{F}_r} = (\theta_0^2\theta_1)^r(F_p) = p^r$ and, for $i \in \Xi_k(p)$,

$$\begin{aligned} F_p|_{\mathcal{F}_i} = (\theta_0^k\theta_1^i)(F_p) &= \left((-1)^{(p-1)/2}\right)^{r-i} p^r \\ &= \begin{cases} p^r, & (k-2i)/p \equiv 0 \bmod 4, \\ (-1)^{(p-1)/2}p^r, & (k-2i)/p \equiv 2 \bmod 4. \end{cases} \end{aligned}$$

Noticing the counting

$$\begin{aligned} \#\{i \in \Xi_k(p) \mid (k-2i)/p \equiv 0 \bmod 4\} &= \left\lfloor \frac{k}{4p} \right\rfloor = n_k^+(p), \\ \#\{i \in \Xi_k(p) \mid (k-2i)/p \equiv 2 \bmod 4\} &= \left\lfloor \frac{k}{4p} + \frac{1}{2} \right\rfloor = n_k^-(p), \end{aligned}$$

the formula (22) then follows. □

2.3. **The cohomology.**

Lemma 2.3.1. *We have the vanishing* $\mathrm{H}^i_{\text{ét},?}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \widetilde{\mathscr{V}_k}) = 0$ *for* $i \neq 1$ *and* $? = \emptyset, \mathrm{c}, \mathrm{mid}$.

Proof. Since $\mathbb{G}_\mathrm{m}$ is an affine curve, we have $\mathrm{H}^i_{\text{ét}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \widetilde{\mathscr{V}_k}) = 0$ for $i \neq 0, 1$ by Artin vanishing. By [Kat88, Thm.11.1], the geometric monodromy group of Kl_2 equals SL_2 over $\overline{\mathbb{Q}}_\ell$, which is connected. Therefore, the geometric monodromy group of the pullback $\widetilde{\mathrm{Kl}_2}$ by the double cover [2] remains equal to SL_2. Since the symmetric power $\widetilde{\mathscr{V}_k}$ of the standard representation $\widetilde{\mathscr{V}_1}$ of SL_2 has no nontrivial invariant, one obtains $\mathrm{H}^0_{\text{ét}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \widetilde{\mathscr{V}_k}) = 0$. The duality pairing (10) for $\mathscr{F} = \widetilde{\mathscr{V}_k}$ then yields the vanishing $\mathrm{H}^i_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k) = 0$ for $i \neq 1$. □

Consequently, one has

$$\dim \mathrm{H}^1_{\text{ét}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \widetilde{\mathscr{V}_k}) = \dim \mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \widetilde{\mathscr{V}_k}) = -\chi(\mathbb{P}^1, j_!\widetilde{\mathscr{V}_k}).$$

By the Grothendieck–Ogg–Shafarevich formula and Lemmas 2.2.1, 2.2.2, one obtains readily

$$\begin{aligned}\chi(\mathbb{P}^1, j_!\widetilde{\mathscr{V}_k}) = -\operatorname{sw}_{\tilde{0}}(\widetilde{\mathscr{V}_k}) - \operatorname{sw}_{\tilde{\infty}}(\widetilde{\mathscr{V}_k}) &= -2\big(\lfloor (k+1)/2 \rfloor - |\Xi_k(p)|\big) \\ &= \begin{cases} -2\big(\frac{k+1}{2} - \big\lfloor \frac{k}{2p} + \frac{1}{2} \big\rfloor\big), & k \text{ odd}, \\ -2\big(\frac{k}{2} - \big\lfloor \frac{k}{2p} \big\rfloor\big), & k \text{ even}. \end{cases}\end{aligned}$$

On the other hand, consider the exact sequence on $\mathbb{P}^1$

$$0 \longrightarrow j_!\widetilde{\mathscr{V}_k} \longrightarrow j_*\widetilde{\mathscr{V}_k} \longrightarrow \widetilde{\mathscr{V}_k}^{I_{\tilde{0}}} \oplus \widetilde{\mathscr{V}_k}^{I_{\tilde{\infty}}} \longrightarrow 0.$$

Taking $\mathbf{R}\Gamma$ and by (12) and Lemma 2.3.1, it yields

$$0 \longrightarrow \widetilde{\mathscr{V}_k}^{I_{\tilde{0}}} \oplus \widetilde{\mathscr{V}_k}^{I_{\tilde{\infty}}} \longrightarrow \mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \widetilde{\mathscr{V}_k}) \longrightarrow \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \widetilde{\mathscr{V}_k}) \longrightarrow 0.$$

Up to this point, we obtain the statements in Theorem 2.1.2 except the formulas for $\det(F_p \mid \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k))$.

The determinant. Due to the existence of $(-1)^{k+1}$-symmetric perfect pairing (11) on the pure $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$-module $\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k)$, we have

$$\det\big(F_p \mid \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k)\big) = \pm p^{\delta(k+1)/2}$$

with positive sign for k even. It remains to determine the sign in the case where k is odd.

Together with (18), we let

$$c^+ = \det(-F_p \mid \mathrm{H}^1_{\text{ét}}(\mathbb{P}^1_{\overline{\mathbb{F}}_p}, j_*\mathscr{V}_k^+)), \quad \tilde{c} = \det(-F_p \mid \mathrm{H}^1_{\text{ét}}(\mathbb{P}^1_{\overline{\mathbb{F}}_p}, j_*\widetilde{\mathscr{V}_k})).$$

By (15), we have the relation

$$\tilde{c} = c^+ \cdot c.$$

The value c^+ is given by $(-1)^{\delta^+} \cdot$ (13) for k odd. So we are reduced to computing $\tilde{c}$.

By the product formula [Lau87, Th.(3.2.1.1)],

$$\tilde{c} = p^{k+1} \prod_{x \in |\mathbb{P}^1|} \varepsilon(\mathbb{P}^1_{(x)}, j_*\widetilde{\mathscr{V}_{k(0)}}, \omega_{(x)})$$

for any nonzero global meromorphism 1-form ω on $\mathbb{P}^1$. Each term in the product in the equality is the local ε-factor, see [Lau87, Th.(3.1.5.4)]. We now take $\omega = \mathrm{d}t^2 = [2]^*\mathrm{d}z$.

Since, on $\mathbb{G}_{\mathrm{m}}$, ω has no pole nor zero and $j_*\widetilde{\mathscr{V}_k}$ is lisse, $\varepsilon(\mathbb{P}^1_{(x)}, j_*\widetilde{\mathscr{V}_k}_{(0)}, \omega_{(x)}) = 1$ for $x \in |\mathbb{G}_{\mathrm{m}}|$. One reduces

$$\tilde{c} = p^{k+1}\varepsilon(\mathbb{P}^1_{(\tilde{0})}, j_*\widetilde{\mathscr{V}_k}_{(\tilde{0})}, \omega_{(\tilde{0})}) \cdot \varepsilon(\mathbb{P}^1_{(\widetilde{\infty})}, j_*\widetilde{\mathscr{V}_k}_{(\widetilde{\infty})}, \omega_{(\widetilde{\infty})}). \tag{23}$$

By [FW10, Lemma 1.2] and Lemma 2.2.1, we have $j_*\widetilde{\mathscr{V}_k}_{(\tilde{0})} \simeq j_*{\mathscr{V}_k^+}_{(0)}$. Therefore we obtain

$$\begin{aligned}\varepsilon(\mathbb{P}^1_{(\tilde{0})}, j_*\widetilde{\mathscr{V}_k}_{(\tilde{0})}, \omega_{(\tilde{0})}) &= \varepsilon(\mathbb{P}^1_{(0)}, j_*{\mathscr{V}_k^+}_{(0)}, \mathrm{d}z^2{}_{(0)}) \\ &= p^{k(k+1)/2}p^{k+1}\varepsilon(\mathbb{P}^1_{(0)}, j_*{\mathscr{V}_k^+}_{(0)}, \mathrm{d}z_{(0)}) \quad ([\text{Lau87, (3.1.5.5)}]) \\ &= -p^{(k+1)^2} \quad ([\text{FW10, Prop.1.4}]). \end{aligned} \tag{24}$$

Here, in the second equality, one uses that $\mathrm{d}z^2 = 2z\mathrm{d}z$ has a simple zero at 0, $\det j_*{\mathscr{V}_k^+}_{(0)} = \overline{\mathbb{Q}}_\ell(-\frac{k(k+1)}{2})$ and the corresponding reciprocity map sends $2z$ to $p^{k(k+1)/2}$.

Recall the local structure in Lemma 2.2.2 of $\widetilde{\mathscr{V}_k}$ at $\widetilde{\infty}$. We have, for $\alpha \in \mathbb{F}_p$, the evaluations of the local conductor ([Lau87, (3.1.5.1)])

$$a(\alpha) := a(\mathbb{P}^1_{(\infty)}, j_*\mathscr{L}_\psi(\alpha t) \otimes \kappa_{2(\infty)}, \omega_{(\infty)}) = \begin{cases} -2, & \alpha = 0, \\ -1, & \alpha \neq 0. \end{cases}$$

One has, for $\mathcal{V}_i = \mathcal{K}_i \otimes \mathcal{F}_{i+1}$,

$$\begin{aligned}\varepsilon(\mathbb{P}^1_{(\infty)}, j_*\mathcal{V}_{i(\infty)}, \omega_{(\infty)}) &= \det(F_p \mid \mathcal{F}_{i+1})^{a(4i-2k)}\varepsilon(\mathbb{P}^1_{(\infty)}, j_*\mathcal{K}_{i(\infty)}, \omega_{(\infty)}) \quad ([\text{Lau87, (3.1.5.6)}]) \\ &= \begin{cases} \pm g^{-2k}\cdot p^{-3}g, & p \mid (k-2i) \\ \pm g^{-k}\cdot p^{-2}, & p \nmid (k-2i) \end{cases} \quad [\text{FW10, Lemma 2.5(iv, v)}].\end{aligned}$$

(The sign can be determined explicitly by the results in [FW10]. It is irrelevant for our purpose due to the exponent 2 in the decomposition (21), so we omit it to simplify the presentation in calculations.) Putting things together with again the identity (19) and setting

$$a_k(p) = \#\Xi_k(p) = \left\lfloor \frac{k}{2p} + \frac{1}{2} \right\rfloor,$$

we obtain

$$\begin{aligned}\varepsilon(\mathbb{P}^1_{(\widetilde{\infty})}, j_*\widetilde{\mathscr{V}_k}_{(\widetilde{\infty})}, \omega_{(\widetilde{\infty})}) &= (g^{-k}p^{-2})^{k+1}(g^{-k+1}p^{-1})^{2a_k(p)} \\ &= \left(\frac{-1}{p}\right)^{(k+1)/2}(p^{-(k+1)/2})^{k+4+2a_k(p)}.\end{aligned} \tag{25}$$

Putting (24) and (25) into (23), we thus have

$$\tilde{c} = \left(\frac{-1}{p}\right)^{(k+1)/2}(-p^{(k+1)/2})^{k-2a_k(p)}$$

and hence the formula $c = (-1)^\delta \cdot (16)$ for $c = \tilde{c}/c^+$ when k is odd.

Remark 2.3.2. One can also obtain the above results more directly by investigating the structures of $\mathscr{V}_k = \kappa_2 \otimes \mathscr{V}_k^+$ based on [FW10] instead of working on $\widetilde{\mathscr{V}_k}$. More details can be found in Appendix A.1.

3. Galois representations

In this section, we construct, for each k, the Galois representation of $\mathbb{Q}$ associated with the twisted moments $\{\mathsf{m}_k(p)\}$ of the Kloosterman sums as a subquotient of the étale cohomology of an explicit affine variety. We extract needed information from the work [FSY22, §§2 and 5.1] in §3.1, which gives detailed properties of various realizations of the motives. We prove the modularity Theorem 1.2.2 in the last subsection §3.2.

3.1. **The motives.** Let $\mathbb{F}$ denote a field. Over $\mathbb{F}$, let $\mathscr{K}$ be the affine hypersurface

$$\mathscr{K} = \Big(\sum_{i=1}^{k} y_i + y_i^{-1} \Big) \subset \mathbb{G}_{\mathrm{m}}^k$$

where $\{y_i\}$ is the Cartesian coordinate of the torus $\mathbb{G}_{\mathrm{m}}^k$. Let $\mathfrak{S}_k$ be the k-th symmetric group, acting on $\mathscr{K}$ by permuting the variables y_i. Let $\mu_2 = \{\pm 1\}$ be the finite group of order 2, acting on $\mathscr{K}$ by $y_i \mapsto \pm y_i$ for all i. The two actions commute and yield an action of $\mathfrak{S}_k \times \mu_2$ on $\mathscr{K}$. Consider the character

$$\chi \colon \mathfrak{S}_k \times \mu_2 \longrightarrow \{\pm 1\},$$

defined to be the product of the sign character on $\mathfrak{S}_k$ and the nontrivial one on μ_2. Let $\chi^+ \colon \mathfrak{S}_k \times \mu_2 \longrightarrow \{\pm 1\}$, which is the sign on $\mathfrak{S}_k$ and trivial on μ_2.

Suppose $\mathbb{F} = \mathbb{F}_p$. In [FSY22, Thm.3.12] concerning the moment $\mathsf{m}_k^+(p)$, it is shown that the étale cohomology $\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k^+)$ is isomorphic to the Tate twist of the χ^+-isotypic part $\mathrm{H}^{k-1}_{\text{ét,c}}(\mathscr{K}_{\overline{\mathbb{F}}_p}, \overline{\mathbb{Q}}_\ell)_{\chi^+}(-1)$ of the étale cohomology of $\mathscr{K}_{\overline{\mathbb{F}}_p}$ under the action of $\mathfrak{S}_k \times \mu_2$, and their weight $(k+1)$ parts also match. Taking the χ-isotypic parts instead in the following, we obtain the corresponding statements for the twisted moments $\mathsf{m}_k(p)$ by the same arguments. Again the choices of χ^+ and χ reflect the decomposition in (9) under a double cover.

Proposition 3.1.1. *Take $\mathbb{F} = \mathbb{F}_p$ with $p \geq 3$. We have*

$$\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k) \simeq \mathrm{H}^{k-1}_{\text{ét,c}}(\mathscr{K}_{\overline{\mathbb{F}}_p}, \overline{\mathbb{Q}}_\ell)_\chi(-1),$$
$$\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k) \simeq \big(\mathrm{gr}^W_{k-1} \mathrm{H}^{k-1}_{\text{ét,c}}(\mathscr{K}_{\overline{\mathbb{F}}_p}, \overline{\mathbb{Q}}_\ell)_\chi \big)(-1).$$

Here W denotes the weight filtration.

Take $\mathbb{F} = \mathbb{Q}$. The above proposition leads us to define the motive over $\mathbb{Q}$ associated with the pure part of $\mathsf{m}_k(p)$ as

$$\mathsf{M}_k = \big(\mathrm{gr}^W_{k-1} \mathrm{H}^{k-1}_{\mathrm{c}}(\mathscr{K})_\chi \big)(-1),$$

the Tate twist of the weight $k-1$ piece of the χ-eigenspace of the (various) middle cohomology with compact support of $\mathscr{K}$. It is of rank

$$\mathrm{rk}\, \mathsf{M}_k = \left\lfloor \frac{k+1}{2} \right\rfloor - \delta_{2+4\mathbb{Z}}(k).$$

There is the natural pairing underlying the Poincaré pairing

$$\mathsf{M}_k \otimes \mathsf{M}_k \longrightarrow \mathrm{H}^2(\mathbb{P}^1)^{\otimes(k+1)} = \mathbb{Q}(-k-1), \tag{26}$$

which is $(-1)^{k+1}$-symmetric and perfect.

The étale realization. For a prime number ℓ, let

$$V_{k,\ell} = \big(\operatorname{gr}^W_{k-1} \mathrm{H}^{k-1}_{\text{ét,c}}(\mathscr{K}_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)_\chi\big)(-1)$$

be the étale realization of M_k, furnished with the continuous action of the Galois group

$$\rho_{k,\ell}\colon \operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}(V_{k,\ell}).$$

The representation is compatible with the pairing induced by (26).

For k odd, we let

$$\Theta_k(p) = \{j \text{ odd positive integer} \mid jp \leq k\}. \tag{27}$$

Let $v_p\colon \mathbb{Z}\setminus\{0\} \to \mathbb{Z}$ be the p-adic valuation. We also set $\Theta_k(p) = \Theta_k^+(p) \sqcup \Theta_k^-(p)$ where

$$\Theta_k^+(p) = \{j \in \Theta_k(p) \mid v_p(j) \text{ odd}\}, \quad \Theta_k^-(p) = \{j \in \Theta_k(p) \mid v_p(j) \text{ even}\}. \tag{28}$$

We mention that there is a bijection $\Xi_k(p) \xrightarrow{\sim} \Theta_k(p)$ given by $i \mapsto (k-2i)/p$ where the set $\Xi_k(p)$ is defined in (20).

Proposition 3.1.2. *Let k be a positive odd integer. For any prime ℓ, the ℓ-adic étale realization $V_{k,\ell}$ of M_k is a quadratic space and yields a continuous representation*

$$\rho_{k,\ell}\colon \operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GO}(V_{k,\ell})$$

satisfying the following properties.

(i) *Let p be an odd prime, $p \neq \ell$. Fix a place of $\overline{\mathbb{Q}}$ above p, and $\operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \subset \operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ the corresponding decomposition subgroup. As a representation of $\operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$, $V_{k,\ell}$ decomposes into an orthogonal sum $M \oplus E$ fulfilling the following conditions.*
- *One has $M \otimes \overline{\mathbb{Q}}_\ell \cong \mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k)$.*
- *E is a direct sum $\bigoplus_{a\in\Theta_k(p)} \varepsilon_a \otimes \chi_{\mathrm{cyc}}^{-(k+1)/2}$ where ε_a is the primitive character associated with the extension $\mathbb{Q}_p(\sqrt{(-1)^{(1+ap)/2}2ap})$ of $\mathbb{Q}_p$ (hence ε_a is unramified precisely when $a \in \Theta_k^+(p)$).*

(ii) *We have*

$$\det \rho_{k,\ell} = \chi_D \cdot \chi_{\mathrm{cyc}}^{-(k+1)^2/4}, \quad D = \begin{cases} (-1)^{(k+3)/4} \cdot 4 \cdot k!!, & k \equiv 1 \bmod 4, \\ (-1)^{(k+1)/4} \cdot k!!, & k \equiv -1 \bmod 4, \end{cases} \tag{29}$$

where χ_D is the quadratic character associated with the extension $\mathbb{Q}(\sqrt{D})$ over $\mathbb{Q}$.

Proposition 3.1.3. *Let k be a positive even integer. For any prime ℓ, the ℓ-adic étale realization $V_{k,\ell}$ of M_k is a symplectic space and yields a continuous representation*

$$\rho_{k,\ell}\colon \operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GSp}(V_{k,\ell}).$$

Let p be an odd prime, distinct from ℓ. Fix a place of $\overline{\mathbb{Q}}$ above p and let I_p be the corresponding inertia subgroup of $\operatorname{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \subset \operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Then for all $\sigma \in I_p$, $(\rho_{k,\ell}(\sigma)-1)^2 = 0$ on $V_{k,\ell}$. Moreover, there is an isotropic subspace U of $V_{k,\ell}$, $\dim U = \lfloor k/2p \rfloor$, such that the induced map $\rho_{k,\ell}(\sigma) - 1\colon V_{k,\ell} \to V_{k,\ell}/U$ is zero for $\sigma \in I_p$ and $V_{k,\ell}^{I_p} = U^\perp$. As representations of $\operatorname{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$, we have

$$V_{k,\ell}^{I_p} \otimes \overline{\mathbb{Q}}_\ell \cong \mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_k)/E, \quad E \cong \overline{\mathbb{Q}}_\ell(-k/2)^{\delta_{2+4\mathbb{Z}}(k)}.$$

The above two propositions are the twisted analogues of the corresponding Thm.5.8 and Thm.5.17 for $\mathsf{m}_k^+(p)$ in [FSY22]. The proofs rely on carefully applying the Picard-Lefschetz formula on (an explicit compactification of) $\mathscr{K}$, regarded as defined over $\mathbb{Z}_p$. This $\mathscr{K}$ has additional ordinary quadratic singularities in its reduction, and yields vanishing cycle classes in $\mathrm{H}^{k-1}_{\text{ét,c}}(\mathscr{K}_{\overline{\mathbb{Q}}_p}, \mathbb{Q}_\ell)$, which are complementary to $\mathrm{H}^{k-1}_{\text{ét,c}}(\mathscr{K}_{\overline{\mathbb{F}}_p}, \mathbb{Q}_\ell)$ in a precise sense. Taking the χ-isotypic parts instead of χ^+, the same arguments loc. cit. verbatim give the proofs for the $\mathsf{m}_k(p)$ case, except the formula (29) for the determinant. The latter is an immediate consequence of Proposition 3.1.1, the determinant formula (16), applying to unramified primes $p > k$, and the Chebotarev density theorem.

Proposition 3.1.4. *The representation $V_{k,\ell}$ is de Rham at ℓ; when k is odd, it is crystalline if ℓ is odd and $\ell > k$; when k is even, it is crystalline if $\ell > k/2$.*

Proof. Note that $V_{2,\ell} = 0$, so the assertion is empty when $k = 2$. Otherwise, this is the corresponding statement of [FSY22, Prop.5.23] for $\mathsf{m}_k^+(p)$, which is proved using the geometry of $\mathscr{K}$ and the p-adic Hodge comparison theorems. One argues verbatim as loc. cit., taking the χ-isotypic parts instead of χ^+. □

The de Rham realization. The de Rham realization of the motive M_k is the middle part of the de Rham cohomology

$$V_{k,\mathrm{dR}} = \big(\mathrm{gr}^W_{k-1}\, \mathrm{H}^{k-1}_{\mathrm{dR,c}}(\mathscr{K}/\mathbb{Q})_\chi\big)(-1).$$

It is a $\mathbb{Q}$-vector space equipped with a Hodge filtration. By [FSY22, Prop.4.21] stating the Hodge numbers for the motives underlying $\widetilde{\mathsf{m}}_k(p)$, and subtracting [FSY22, Thm.1.8] for the Hodge numbers of $\mathsf{m}_k^+(p)$, we obtain the Hodge numbers of $V_{k,\mathrm{dR}}$ below. We remark that, unlike the étale realizations $V_{k,\ell}$ where the detailed descriptions as Galois representations are revealed through the geometry of $\mathscr{K}$, the Hodge structures $V_{k,\mathrm{dR}}$ are investigated mainly through an irregular connection on $\mathbb{G}_\mathrm{m}$ over $\mathbb{C}$, which is the complex counterpart of $\mathscr{V}_k$, via irregular Hodge theory (also see [Chu24] for the period realization of the connection). The weight part of the statement can also be read off from Theorem 1.2.1.

Proposition 3.1.5. *Let $V'_{k,\mathrm{dR}} = \mathrm{H}^{k-1}_{\mathrm{dR,c}}(\mathscr{K}/\mathbb{Q})_\chi(-1)$; it has (Hodge) weights $\le k+1$ and contains $V_{k,\mathrm{dR}}$ as its top weighted quotient.*

(i) *When $k \not\equiv 2 \bmod 4$, we have $V'_{k,\mathrm{dR}} \xrightarrow{\sim} V_{k,\mathrm{dR}}$ and they are of dimension $\lfloor (k+1)/2 \rfloor$. The Hodge numbers $h^{p,k+1-p} = \dim \mathrm{gr}^p_F V'_{k,\mathrm{dR}} = 1$ if $p = 2i-1$ or $k+2-2i$ for $1 \le i \le \lfloor (k+3)/4 \rfloor$, and vanish otherwise.*

(ii) *If $k \equiv 2 \bmod 4$, then $\dim V'_{k,\mathrm{dR}} = k/2$ and $\dim V_{k,\mathrm{dR}} = (k-2)/2$. The Hodge numbers $h^{p,q} = \dim \mathrm{gr}^W_{p+q}\, \mathrm{gr}^p_F\, V'_{k,\mathrm{dR}}$ vanish except in the following cases, in which $h^{p,q} = 1$,*

$$\begin{cases} p \text{ or } q = 2i-1 \text{ for } 1 \le i \le (k-2)/4, \text{ and } p+q = k+1, \\ p = q = k/2. \end{cases}$$

Corollary 3.1.6. *The representation $\rho_{k,\ell}$ is potentially automorphic.*

Proof. As in [FSY22, §5.3.3], we apply the main automorphy result of [PT15] to the weakly compatible system $\{\rho_{k,\ell}\}_\ell$ for k fixed. By Proposition 3.1.4, $\rho_{k,\ell}$ is de Rham above ℓ, and is crystalline when ℓ is large enough. We have the three properties.

(i) (Purity) For p odd and $p > k$ if k is odd (resp., $p > k/2$ if k is even), $\rho_{k,\ell}$ is unramified at p, and the roots of $\rho_{k,\ell}(\mathrm{Frob}_p)$ of the geometric Frobenius are Weil $p^{(k+1)}$-numbers.
(ii) (Regularity) The nonvanishing Hodge numbers $h^{i,j}$ of (the underlying motive of) $\rho_{k,\ell}$ are equal to one.
(iii) (Odd essential self-duality) $\rho_{k,\ell}$ is $(-1)^{k+1}$-symmetric with similitude character $\chi_{\mathrm{cyc}}^{-k-1}$; the determinant $\{\det \rho_{k,\ell}\}_\ell$ also forms a weakly compatible system.

Therefore, by [PT15, Thm.A], $\rho_{k,\ell}$ is potentially automorphic. □

Recall the polynomial $Z_k(p;T)$, the set $\Theta_k(p)$ and the numbers $n_k^{\pm}(p)$ defined in (17), (27) and (14), respectively. Let $\mathbb{B}_{\mathrm{crys}}$ and $\mathbb{B}_{\mathrm{st}}$ be Fontaine's p-adic crystalline and semistable period rings over $\mathbb{Q}_p$. They are filtered rings with an absolute Frobenius (semilinear) automorphism φ. We have the following strengthening of Proposition 3.1.4.

Corollary 3.1.7. *Let p be an odd prime. Choose a place of $\overline{\mathbb{Q}}$ above p, and consider $V_{k,p}$ as a representation of the corresponding decomposition subgroup $\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$.*

(i) *Suppose k is odd. Then $V_{k,p}$ is crystalline over $L = \mathbb{Q}_p(\sqrt{-p})$ and*

$$\det\left(1-\varphi T \mid (V_{k,p}\otimes \mathbb{B}_{\mathrm{crys}})^{\mathrm{Gal}(\overline{\mathbb{Q}}_p/L)}\right) = Z_k(p;T)\prod_{a\in\Theta_k(p)}\left(1-\left(\frac{(-1)^{(1+ap)/2}2a'}{p}\right)p^{\frac{k+1}{2}}T\right) \tag{30}$$

where $a' = ap^{-v_p(a)}$ is the prime-to-p part of a. If $p > k$, $V_{k,p}$ is crystalline over $\mathbb{Q}_p$ and one can replace L by $\mathbb{Q}_p$ in the equation.

(ii) *Suppose k is even. Then $V_{k,p}$ is semistable over $\mathbb{Q}_p$ and*

$$\begin{aligned}\det\left(1-\varphi T \mid (V_{k,p}\otimes \mathbb{B}_{\mathrm{st}})^{\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)}\right) &= Z_k(p;T)\left(1-p^{\frac{k}{2}}T\right)^{-\delta_{2+4\mathbb{Z}}(k)}\\ &\quad\cdot\left(1-p^{\frac{k+2}{2}}T\right)^{n_k^+(p)}\left(1-(-1)^{\frac{p-1}{2}}p^{\frac{k+2}{2}}T\right)^{n_k^-(p)}.\end{aligned} \tag{31}$$

If $p > k/2$, $V_{k,p}$ is crystalline over $\mathbb{Q}_p$ and one can replace $\mathbb{B}_{\mathrm{st}}$ by $\mathbb{B}_{\mathrm{crys}}$ in the equation.

In particular, the p-adic Newton polygons of the polynomials (30) and (31) are above the Hodge polygon of $V_{k,\mathrm{dR}}$, and the polygons have the same endpoints.

Proof. (Cf. [FSY22, Rem.5.41].) A consequence of the potential automorphy Corollary 3.1.6 is that the system $\{\rho_{k,\ell}\}_\ell$ is indeed strictly compatible ([PT15, Cor.2.2(ii)]). Thus the Weil-Deligne representation (r_ℓ, N) of $\mathbb{Q}_p$ associated with $V_{k,\ell}$ for $\ell \neq p$ (cf. [FSY22, §5.3.1]) equals that of $V_{k,p}$ constructed via the p-adic period rings (Fontaine's functors).

Suppose k is odd. By Proposition 3.1.2(i), $V_{k,\ell}$ is unramified over L and $\det(1-\mathrm{Frob}_p T \mid V_{k,\ell})$ equals the right-hand side of (30). Here we regard Frob_p as in the quotient of $\mathrm{Gal}(\overline{\mathbb{Q}}_p/L)$.

Assume k is even. By Proposition 3.1.3, the associated Weil representation r_ℓ is unramified. Let $\{e_i\}$ be a basis of the subspace $U \subset V_{k,\ell}^{I_p}$ in Proposition 3.1.3. Then there are elements $\{e_i'\}$, inducing a basis of $V_{k,\ell}/U^\perp$, such that $\mathrm{N}e_i' = e_i$. Since $\mathrm{N}\circ\mathrm{Frob}_p = p\mathrm{Frob}_p\circ\mathrm{N}$ and

$$\det(1-\mathrm{Frob}_p T \mid U) = (1-p^{k/2}T)^{n_k^+(p)}(1-(-1)^{(p-1)/2}p^{k/2}T)^{n_k^-(p)}$$

by Theorem 2.1.2(ii)(a), the equality (31) follows.

On the other hand, the associated filtered φ-module of $V_{k,p}$ are (weakly) admissible. Therefore the attached Newton polygon lies above the Hodge polygon, and the two have the same endpoints.

We remark that one can deduce weaker statements using rigid cohomology and p-adic Hodge theory more directly, cf. [FSY22, Prop.5.23, Cor.5.27]. □

3.2. **Modularity.** Here we prove Theorem 1.2.2. In the cases $k = 3, 4, 6$, the associated Galois representations $\rho_{k,\ell}$ of M_k are of rank 2. The modularity is obtained similarly to that in [Ser87, §4.8]. For the rank 3 representation $\rho_{5,\ell}$, we argue as in [Yun15, §4.7].

In all cases, the exponents c of 2 in the level N of the associated cuspidal modular forms are at most 8 by the last paragraph of [Ser87, p.216].

Let $\chi_{\mathrm{cyc},\ell}$ denotes the ℓ-adic cyclotomic character of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$.

The case $k = 3$. Due to the existence of the symmetric perfect pairing (26), the orthogonal representation $\rho_{3,\ell} \otimes \chi_{\mathrm{cyc},\ell}$ is modular (with CM) of Serre weight 3 by [Liv95, Thm.1.3]. Let f be the corresponding cusp form. By a theorem of Fontaine ([Edi92, Thm.2.6]), the reduction $\overline{\rho}_{3,\ell}\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}(\mathbb{F}_\ell)$ mod ℓ of $\rho_{3,\ell}$ is absolutely irreducible for those ℓ such that the Fourier coefficient $a_f(\ell) = 0$ (there are infinitely many since f has CM). Now we apply the strong form of Serre's modularity conjecture to pin down f. By the recipe in [Ser87], we look for a normalized Hecke eigenform with coefficients in $\mathbb{Q}$ among those of weight 3 and level $2^c \cdot 3$ for some $0 \le c \le 8$. In `LMFDB`, it results in 8 matches. The conditions

$$a_f(7) = -\mathsf{m}_3(7)/7 = -2, \quad a_f(13) = -\mathsf{m}_3(13)/13 = -22$$

($a_f(5) = a_f(11) = 0$) single out one such form $f_{48.3.e.a}$ (which has CM by $\mathbb{Q}(\sqrt{-3})$).

Remark 3.2.1. Let $g = f_{12.3.c.a} = f_{48.3.e.a} \otimes \left(\frac{-4}{\bullet}\right)$. It is twist minimal and satisfies

$$a_g(p) = (-1)^{(p-1)/2}\mathsf{m}_3(p)/p.$$

The equality is the statement in [Eva10, p.523].

The case $k = 4$. Applying [Ser87, Th.6] directly to the symplectic representation $\rho_{4,\ell} \otimes \chi_{\mathrm{cyc},\ell}$, we look for a normalized Hecke eigenform among those of weight 4, level 2^c for some $0 \le c \le 8$, and with coefficients in $\mathbb{Q}$. In `LMFDB`, it results in 22 matches. The conditions

$$a_f(3) = -\mathsf{m}_4(3)/3 = -4, \quad a_f(5) = -\mathsf{m}_4(5)/5 = -2$$

single out one such form $f_{8.4.a.a}$ We remark that this form is twist minimal.

The case $k = 6$. Applying directly [Ser87, Th.6] to the symplectic representation $\rho_{6,\ell} \otimes \chi_{\mathrm{cyc},\ell}$, we look for a normalized Hecke eigenform among those of weight 6, level $2^c \cdot 3, 0 \le c \le 8$, with coefficients in $\mathbb{Q}$. In `LMFDB`, it results in 48 matches. The conditions

$$a_f(5) = -\mathsf{m}_6(5)/5 - 5^2 = -34, \quad a_f(7) = -\mathsf{m}_6(7)/7 - 7^2 = -240$$

single out one such form $f_{24.6.a.a}$. We remark that this form is twist minimal.

The case $k = 5$. Consider the Galois representations $\rho_{5,\ell}$ constructed in Theorem 3.1.2. Let

$$r_\ell = \rho_{5,\ell} \otimes \chi_{\mathrm{cyc},\ell}^3 \otimes \left(\frac{15}{\bullet}\right)\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{SO}(V_{5,\ell}). \tag{32}$$

It satisfies the following properties:

- The representations r_ℓ is unramified at p for every prime $p \neq 2, 3, 5, \ell$. In this case, the trace of the geometric Frobenius is given by

$$\mathrm{tr}(r_\ell(\mathrm{Frob}_p)) = \left(\frac{15}{p}\right)\frac{-\mathsf{m}_5(p)}{p^3}. \tag{33}$$

- It is tamely ramified at $p = 3, 5$.
- It is de Rham at $p = \ell$, and has Hodge-Tate weight $(2, 0, -2)$. It is indeed crystalline if $\ell > 5$.
- The image $r_\ell(\iota)$ of the complex conjugation ι is conjugate to $\mathrm{diag}(1, -1, -1)$.

Fix the choices of the identifications

$$\mathrm{SO}(V_{5,\ell} \otimes_{\mathbb{Q}_\ell} \overline{\mathbb{Q}}_\ell) \simeq \mathrm{SO}(3)(\overline{\mathbb{Q}}_\ell) \simeq \mathrm{PGL}_2(\overline{\mathbb{Q}}_\ell).$$

Let

$$r'_\ell \colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{PGL}_2(\overline{\mathbb{Q}}_\ell)$$

be the *same* representation as r_ℓ under the identifications. In particular,

$$r'_\ell(\iota) \sim \begin{pmatrix} 1 & & \\ & & \\ & & -1 \end{pmatrix}. \tag{34}$$

Lemma 3.2.2. *Suppose $\ell > 5$. For $p = 3, 5$, the representation r'_ℓ is tamely ramified at p, and it sends a generator of the tame quotient I_p^t of the inertia subgroup $I_p \subset \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ to a matrix conjugate to $\mathrm{diag}(1, -1)$ (up to scaling). Moreover, the image $r'_\ell(\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)) \subset \mathrm{PGL}_2(\overline{\mathbb{Q}}_\ell)$ is conjugate to a subgroup of the diagonal matrices (up to scaling).*

Proof. Since r_ℓ is tamely ramified at odd primes, the same holds for r'_ℓ. For $p = 3, 5$, choose a topological generator σ of I_p^t. Since $\dim_{\mathbb{Q}_\ell} V_{5,\ell}^{I_p} = 2$, $\rho_\ell(\sigma)$ is similar to $\mathrm{diag}(1, 1, -1)$. As a result, $r_\ell(\sigma)$ is similar to $\mathrm{diag}(-1, -1, 1)$ in $\mathrm{SO}(3)(\overline{\mathbb{Q}}_\ell)$, and hence $r'_\ell(\sigma)$ is similar to $\mathrm{diag}(1, -1)$ in $\mathrm{PGL}_2(\overline{\mathbb{Q}}_\ell)$. In the following, we choose a frame such that $r'_\ell(\sigma) = \mathrm{diag}(1, -1)$ (up to scaling).

Let $\phi_p \in \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ be a lifting of the geometric Frobenius. We have $\phi_p^{-1}\sigma\phi_p = \sigma^p$ in I_p^t, and hence $r'_\ell(\phi_p)$ is equal to $\left(\begin{smallmatrix} a & 0 \\ 0 & d \end{smallmatrix}\right)$ or $\left(\begin{smallmatrix} 0 & b \\ c & 0 \end{smallmatrix}\right)$. In these two cases, $r_\ell(\phi_p)$ are then similar to

$$\begin{pmatrix} ad^{-1} & & \\ & a^{-1}d & \\ & & 1 \end{pmatrix}, \quad \begin{pmatrix} & bc^{-1} & \\ b^{-1}c & & \\ & & -1 \end{pmatrix},$$

respectively, in which the actions on the two-dimensional subspace $(\overline{\mathbb{Q}}_\ell^3)^{\sigma+1}$ occor in the upper-left parts. Observe that $\det(1 - F_p T \mid \mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathscr{V}_5)^{I_p}) = Z_5(p; T)$ are

$$1 + 6T + 729T^2 \quad \text{and} \quad 1 + 150T + 15625T^2$$

for $p = 3$ and 5, respectively. This implies in particular that $\mathrm{tr}(r_\ell(\phi_p) \mid (\overline{\mathbb{Q}}_\ell^3)^{\sigma+1}) \neq 0$. Hence, $r'_\ell(\phi_p) = \mathrm{diag}(a, d)$, and the proof is completed. □

Proposition 3.2.3 (Cf. [Yun15, Lemma 4.7.6])**.** *For every prime $\ell > 5$, there exists a lifting*

$$\widetilde{r}_\ell \colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_2(\overline{\mathbb{Q}}_\ell)$$

of $r'_\ell \colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{PGL}_2(\overline{\mathbb{Q}}_\ell)$ satisfying the following properties:

- *For prime $p \neq 2, 3, 5, \ell$, the representation $\widetilde{r}_\ell$ is unramified at p.*
- *For prime $p = 3, 5$, the representation $\widetilde{r}_\ell$ is tamely ramified at p, and it sends a generator of I_p^t to a matrix conjugate to $\mathrm{diag}(1, -1)$.*
- *$\widetilde{r}_\ell$ is crystalline at ℓ and its Hodge-Tate weight equals $(2, 0)$.*
- $\det \widetilde{r}_\ell = \left(\frac{-15}{\bullet}\right) \otimes \chi_{\mathrm{cyc},\ell}^{-2}$.

We then have $r_\ell = \mathrm{Sym}^2 \widetilde{r}_\ell \otimes \det^{-1} \widetilde{r}_\ell$ *over* $\overline{\mathbb{Q}}_\ell$.

Proof. By [Pat15, Prop.5.5], there exists a lifting $\Phi\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\overline{\mathbb{Q}}_\ell)$ of r'_ℓ, which is unramified almost everywhere and is de Rham at ℓ.

For $p \neq 2, 3, 5, \ell$, we have $\Phi|_{I_p} = c_p \mathrm{Id}$ for some character c_p of I_p since $r'_\ell|_{I_p}$ is trivial. Note that c_p are trivial for almost all p since Φ is unramified almost everywhere.

For $p = 3, 5$, the representation $\Phi|_{\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)}$ is isomorphic to $d_p \oplus d'_p$ by Lemma 3.2.2. The restrictions of d_p and d'_p at the wild inertia group are the same since r'_ℓ is tamely ramified. We denote c_p the restriction of d_p to the inertia group.

Since r'_ℓ is crystalline and the lifting Φ is de Rham at ℓ, by [Con11, Cor.6.7], there exists a crystalline lifting ϕ of $r'_\ell|_{\mathrm{Gal}(\overline{\mathbb{Q}}_\ell/\mathbb{Q}_\ell)}$. We may assume that ϕ has Hodge-Tate weight $(2, 0)$ after a suitable Tate twist. Then $\Phi|_{\mathrm{Gal}(\overline{\mathbb{Q}}_\ell/\mathbb{Q}_\ell)} = \phi \cdot d_\ell$ for a continuous character $d_\ell\colon \mathrm{Gal}(\overline{\mathbb{Q}}_\ell/\mathbb{Q}_\ell) \to \overline{\mathbb{Q}}_\ell^\times$. Take $c_\ell = d_\ell|_{I_\ell}$.

At $p = 2$, identify I_2 with $\{\pm 1\} \times \mathbb{Z}_2$ where $\{\pm 1\} = \mathrm{Gal}(\mathbb{Q}_2(\sqrt{-1})/\mathbb{Q}_2)$. The restriction $\Phi|_{I_2}$ is quasi-unipotent, and hence $\det \Phi$ factors through a finite quotient $\{\pm 1\} \times \mathbb{Z}/2^a$. Pick an $\alpha \in \overline{\mathbb{Q}}_\ell^\times$ such that $\det \Phi|_{I_2}(1, 1) = \alpha^2$. Let $c_2 : I_2 \to \overline{\mathbb{Q}}_\ell^\times$ given by $c_2|_{\{\pm 1\}} = 1$ and $c_2(1, 1) = \alpha$ so that $c_2^{-2} \det \Phi|_{I_2}$ factors through $\{\pm 1\}$.

Let $c\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})^{\mathrm{ab}} = \prod_p \mathbb{Z}_p^\times \to \overline{\mathbb{Q}}_\ell^\times$ be the product of all c_p via the identification $I_p = \mathbb{Z}_p^\times$ (permuting p^m-th roots of unity for $m \geq 1$). This c is well-defined, as c_p is trivial for almost every p. Then $\widetilde{r}_\ell = \Phi \otimes c^{-1}$ is a lifting satisfying the first three listed conditions with the twist of the determinant $(\det \widetilde{r}_\ell) \otimes \chi_{\mathrm{cyc},\ell}^2$ equal to either $\left(\frac{-15}{\bullet}\right)$ or $\left(\frac{60}{\bullet}\right)$ since it has conductor 15 or 60. Since $\widetilde{r}_\ell$ is odd by (34), one has $(\det \widetilde{r}_\ell) \otimes \chi_{\mathrm{cyc},\ell}^2 = \left(\frac{-15}{\bullet}\right)$. □

Consider the reduction mod ℓ

$$\overline{\widetilde{r}}_\ell\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_2(\overline{\mathbb{F}}_\ell)$$

of $\widetilde{r}_\ell$. It is well-defined up to semisimplification.

Lemma 3.2.4. *The reduction* $\overline{\widetilde{r}}_\ell$ *is irreducible for* $\ell \gg 1$.

Proof. (Cf. [Ser87, (4.8.9)], [Kis07, Thm.1.2.1], [Yun15, Lemma 4.7.2] and their proofs.) Suppose, on the contrary, that there are infinitely many primes ℓ such that $\overline{\widetilde{r}}_\ell$ is not irreducible. So, for these ℓ, there are characters $\psi_1, \psi_2\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \overline{\mathbb{F}}_\ell^\times$ such that $\overline{\widetilde{r}}_\ell \simeq \psi_1 \oplus \psi_2$ up to semisimplification. Then, for the reduction $\overline{r}_\ell$ of r_ℓ in (32), we have $\overline{r}_\ell \simeq \psi_1\psi_2^{-1} \oplus 1 \oplus \psi_1^{-1}\psi_2$. Let N be the conductor of r_ℓ. Since r_ℓ is de Rham at ℓ, there is a Dirichlet character $\chi_0\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \overline{\mathbb{Q}}^\times \subset \overline{\mathbb{Q}}_\ell^\times$ of conductor dividing N such that $\psi_1\psi_2^{-1} = \overline{\chi}$ where $\overline{\chi}$ is the reduction of $\chi = \chi_{\mathrm{cyc},\ell}^{-2}\chi_0$ (after exchanging ψ_1, ψ_2 if needed). In particular, by the evaluation of $\rho_{5,\ell} = r_\ell \chi_{\mathrm{cyc},\ell}^{-3}\left(\frac{15}{\bullet}\right)$ at Frob_p for $p > 5, p \neq \ell$, one has

$$-\left(\frac{15}{p}\right)\mathsf{m}_5(p) \equiv p^3\left(p^2\chi_0(p) + 1 + p^{-2}\chi_0^{-1}(p)\right) \bmod \ell. \tag{35}$$

Since the congruence holds for arbitrarily large ℓ and the involved characters χ_0 have bounded conductors, there exists a fixed such χ_0 making (35) an equality in the ring of integers of a fixed cyclotomic field (adding values of χ_0 to $\mathbb{Q}$).

On the other hand, the integer $\mathsf{m}_5(p)$ is in the range $[-3p^3, 3p^3]$ since $\rho_{5,\ell}$ is of rank 3 and weight 6. However, one has

$$|p^5\chi_0(p) + p^3 + p\chi_0^{-1}(p)| \geq p^5 - p^3 - p,$$

and the right-hand side is greater than $3p^3$ for $p \geq 5$, yielding a contradiction. □

Lemma 3.2.5. *For $p \equiv 5 \bmod 6$, we have*

$$\frac{-\mathsf{m}_5(p)}{p} + \Big(\frac{15}{p}\Big)p^2 \quad \in 5\mathbb{Z}.$$

Furthermore, up to $p = 2357$ (the 350th prime), the values $\mathsf{m}_5(p)$ are of the form

$$\Big(\frac{-5}{p}\Big)\frac{-\mathsf{m}_5(p)}{p} + \Big(\frac{-3}{p}\Big)p^2 = \begin{cases} m_p^2, & \text{if } p \equiv 1 \bmod 6, \\ -5n_p^2, & \text{if } p \equiv 5 \bmod 6, \end{cases}$$

where m_p and $n_p \in \mathbb{Z}$.

Proof. Immediately from the definition, one has

$$\mathrm{Kl}_2^{\mathrm{Sym}^5}(a;p) = \mathrm{Kl}_2^{\mathrm{Sym}^1}(p;a)^5 - 4p\,\mathrm{Kl}_2^{\mathrm{Sym}^3}(a;p) - 5\,\mathrm{Kl}_2^{\mathrm{Sym}^1}(p;a)p^2.$$

By the direct evaluation (3) of $\mathsf{m}_1(p)$, the modularity of $\mathsf{m}_3(p)$ in (4) and that $f = f_{48.3.\mathrm{e.a}}$ has CM by $\mathbb{Q}(\sqrt{-3})$, we can write

$$\begin{aligned} \mathsf{m}_5(p) &= -\sum_{a\in\mathbb{F}_p^\times}\Big(\frac{a}{p}\Big)\mathrm{Kl}_2(p;a)^5 + 4p^2\cdot a_f(p) + 5\Big(\frac{-1}{p}\Big)p^3 \\ &= -\sum_{a\in\mathbb{F}_p^\times}\Big(\frac{a}{p}\Big)\mathrm{Kl}_2(p;a)^5 + 5\Big(\frac{-1}{p}\Big)p^3, \end{aligned}$$

for $p \equiv 5 \bmod 6$. Observe that

$$\frac{-\mathsf{m}_5(p)}{p} \equiv \frac{1}{p}\sum_{a\in\mathbb{F}_p^\times}\Big(\frac{a}{p}\Big)\mathrm{Kl}_2(p;a)^5 \equiv \frac{1}{p}\sum_{a\in\mathbb{F}_p^\times}\Big(\frac{a}{p}\Big)\mathrm{Kl}_2(p;a) = \Big(\frac{-1}{p}\Big) \equiv \Big(\frac{-5}{p}\Big)p^2 \bmod 5,$$

since $p^2 \equiv \left(\frac{5}{p}\right) \bmod 5$.

The second assertion can be derived from common software. □

We are now ready to prove the modularity of m_5. For a given prime $\ell > 5$, let $r_\ell = \rho_{5,\ell} \otimes \chi_{\mathrm{cyc},\ell}^3 \otimes \left(\frac{15}{\bullet}\right)$. Then $r_\ell = \mathrm{Sym}^2 \widetilde{r}_\ell \otimes \det^{-1}\widetilde{r}_\ell$ for a degree-2 representation $\widetilde{r}_\ell$ as in Proposition 3.2.3.

Let $\bar{\widetilde{r}}_\ell$ be the mod ℓ representation of $\widetilde{r}_\ell$, well-defined up to semisimplification. We may assume that $\bar{\widetilde{r}}_\ell$ is absolutely irreducible by choosing $\ell \gg 0$, by Lemma 3.2.4. The Artin conductor of $\widetilde{r}_\ell$ is $2^c \cdot 3 \cdot 5$, for some $c \in \mathbb{Z}_{\geq 0}$, by Proposition 3.2.3. By Proposition 3.1.5, we deduce that the Serre weight of $\widetilde{r}_\ell$ is 3. The assertion of Serre's modularity conjecture tells us that there exists a newform f of level $2^c \cdot 15$ for some $0 \leq c \leq 8$, weight 3, character ϵ_f with q-expansion $\sum a_f(n)q^n$ and associated Galois representation ρ_f such that the reduction mod ℓ representation of ρ_f is isomorphic to $\bar{\widetilde{r}}_\ell$.

By Proposition 3.2.3, the character ϵ_f is induced from $\left(\frac{-15}{\bullet}\right)$. Therefore, the target Hecke eighenform f satisfies that

$$a_f(p)^2 - \left(\frac{-15}{p}\right)p^2 = \left(\frac{-15}{p}\right)p^2 \cdot \mathrm{tr}(r_\ell(\mathrm{Frob}_p)) = \left(\frac{-1}{p}\right)\frac{-\mathsf{m}_5(p)}{p},$$

for prime $p \geq 7$. Further by Lemma 3.2.5 for the case of $p \equiv 5 \bmod 6$, we must have

$$\left(\frac{a_f(p)}{\sqrt{-5}n_p}\right)^2 = \left(\frac{-15}{p}\right)\left(\frac{-3}{p}\right) = \left(\frac{5}{p}\right), \quad \text{for some } n_p \in \mathbb{Z}.$$

Also, for $p \equiv 1 \bmod 6$ with $p \leq 2357$, we have

$$\left(\frac{a_f(p)}{m_p}\right)^2 = \left(\frac{-15}{p}\right)\left(\frac{-3}{p}\right) = \left(\frac{5}{p}\right), \quad \text{for some } m_p \in \mathbb{Z}.$$

So $a_f(p) = \pm m_p, \pm m_p i, \pm\sqrt{5}n_p, \pm\sqrt{-5}n_p$. Since the Sturm bound for the space of modular form of level $2^8 \cdot 15$ with a given character is 2304, we conclude that the coefficient field $K_f := \mathbb{Q}(a_f(n) : n \geq 1)$ is $\mathbb{Q}(\sqrt{5}, i)$.

Taking f to be any of the normalized newform in the space $S_3\big(60, \left(\frac{-15}{\bullet}\right)\big)$ with $K_f = \mathbb{Q}(\sqrt{5}, i)$, one can check the validity of the identity numerically

$$a_f(p)^2 = \left(\frac{-1}{p}\right)\frac{-\mathsf{m}_5(p)}{p} + \left(\frac{-15}{p}\right)p^2 \quad \text{for primes } 7 \leq p \leq 2309,$$

and this confirms the claimed modularity for m_5.

Remark 3.2.6. Suppose f satisfies the equality (5) in Theorem 1.2.2. Let $g = f \otimes \psi$ be the twist of f by a character ψ. Then the coefficients $a_g(p) = \psi(p)a_f(p)$ satisfy a similar equality, like (36) below. In this sense, the modular form f is the one with minimal level that interprets $\mathsf{m}_5(p)$ in terms of (the symmetric squares of) modular forms. We refer the computation of the level of a twist to [BBB+21, Lemma 11.2.1].

In particular, let g be one of the forms $f_{300.3.g.e}$ and $f_{300.3.g.h}$ in `LMFDB`, which are the twists of $f_{60.3.b.a}$ by the characters $\chi = \chi_{5.c}$ or $\chi_{15.e}$. We have $\chi^2(n) = \left(\frac{5}{n}\right)$, and hence one has

$$\left(\frac{-3}{p}\right)a_g(p)^2 = p^2\left[\left(\frac{15}{p}\right)\frac{-\mathsf{m}_5(p)}{p^3} + 1\right] \tag{36}$$

for prime $p > 5$. On the other hand, we have $\left(\frac{15}{p}\right)\frac{-\mathsf{m}_5(p)}{p^3} = \mathrm{tr}(r_\ell(\mathrm{Frob}_p)) \in \mathbb{Q}$ for $r_\ell(\mathrm{Frob}_p) \in \mathrm{SO}(V_{5,\ell})$ in (33). Now choosing a basis of $(V_{5,\ell})_{\overline{\mathbb{Q}}_\ell}$ and taking an isomorphism $\overline{\mathbb{Q}}_\ell \cong \mathbb{C}$ as abstract fields, we regard $r_\ell(\mathrm{Frob}_p)$ as an element in $\mathrm{SO}(3, \mathbb{C})$. Regarding traces, we may assume that $r_\ell(\mathrm{Frob}_p)$ is semisimple. Then $r_\ell(\mathrm{Frob}_p)$ lies in a compact subgroup of $\mathrm{SO}(3, \mathbb{C})$ since the eigenvalues have absolute value one. Therefore $r_\ell(\mathrm{Frob}_p)$ is conjugate to an element in the compact form $\mathrm{SO}(3)$. For $T \in \mathrm{SO}(3)$, one has $\mathrm{tr}(T) = 1 + 2\cos\theta \geq -1$ since every nontrivial rotation T on $\mathbb{R}^3$ is determined by its axis and angle of rotation $\theta \in \mathbb{R}/2\pi\mathbb{Z}$. Consequently, one can replace the term on the left-hand side of (36) by $|a_g(p)|^2$, which recovers Evans's original formulation in [Eva10, p.524]. The positivity is consistent with the observation that $a_n(g) \in \mathbb{Z}$ when $n \equiv 1 \bmod 6$ and $a_n(g) \in \mathbb{Z} \cdot \sqrt{-5}$ when $n \equiv 5 \bmod 6$ in `LMFDB`.

Appendix A. Additional materials

A.1. **Direct approaches.** Deligne [Del77] and Katz [Kat88] had given cohomological interpretation of the Kloosterman sums. Let p, ℓ be different prime numbers and $\overline{\mathbb{Q}}_\ell$ be an algebraic closure of the field of ℓ-adic numbers $\mathbb{Q}_\ell$. Then the nontrivial additive character $\psi\colon \mathbb{F}_p \to \mathbb{C}^\times$ could be regard as character taking value in $\overline{\mathbb{Q}}_\ell$ by choosing a primitive p-th root of unity in $\overline{\mathbb{Q}}_\ell$. Denote the Artin-Schreier sheaf as $\mathscr{L}_\psi$, which is a rank one ℓ-adic lisse étale sheaf on $\mathbb{A}^1_{\mathbb{F}_p}$ with trace function $z \mapsto \psi(\mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_p}(z))$. Now, consider the following diagram:

(37)
$$\begin{array}{ccccc} & & \mathbb{G}^2_{m,\mathbb{F}_p} & & \\ & \overset{\gamma}{\swarrow} & & \overset{\sigma}{\searrow} & \\ \mathbb{G}_{m,\mathbb{F}_p} & & & & \mathbb{A}^1_{\mathbb{F}_p} \end{array}$$

where σ and γ are sum and product of coordinates respectively, i.e., if the coordinates of $\mathbb{G}^2_{m,\mathbb{F}_p}$ are (z,x), then $\sigma(z,x) = z+x$ and $\gamma(z,x) = zx$. The Kloosterman sheaf Kl_2 is defined as:

$$\mathrm{Kl}_2 = \mathrm{R}^1\gamma_!\sigma^*\,\mathscr{L}_\psi\,.$$

The Kloosterman sheaf is a lisse etale sheaf of rank two on $\mathbb{G}_{m,\mathbb{F}_p}$ with trace function $a \mapsto -\mathrm{Kl}_2(q;a)$ if $a \in \mathbb{G}_m(\mathbb{F}_q)$. Moreover, Kl_2 is tame at 0 and totally wild at ∞.

By changing variables of $\mathbb{G}^2_{m,\mathbb{F}_p}$ by $(z,x) \mapsto (z/x, x)$, the diagram (37) could be regard as:

$$\begin{array}{ccccc} & & \mathbb{G}^2_{m,\mathbb{F}_p} & & \\ & \overset{\pi}{\swarrow} & & \overset{f}{\searrow} & \\ \mathbb{G}_{m,\mathbb{F}_p} & & & & \mathbb{A}^1_{\mathbb{F}_p} \end{array}$$

where $\pi(z,x) = z$, $f(z,x) = x + z/x$ and $\mathrm{Kl}_2 \simeq \mathrm{R}^1\pi_! f^*\,\mathscr{L}_\psi$.

Let $[2]\colon \mathbb{G}_{m,\mathbb{F}_p} \to \mathbb{G}_{m,\mathbb{F}_p}$, $t \mapsto t^2$ be the double cover, and $\widetilde{\mathrm{Kl}_2} = [2]^*\,\mathrm{Kl}_2$. Furthermore, denote $\left(\frac{\cdot}{p}\right)$ as the sheaf of Legendre's symbol on $\mathbb{G}_{m,\mathbb{F}_p}$.

First, we recall some vanishing results.

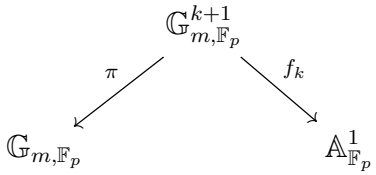
Proposition A.1.1. *The following cohomology groups vanish.*

(i) $\mathrm{H}^i_{\text{ét},?}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, \mathrm{Sym}^k\,\mathrm{Kl}_2) = 0$ *for all* $? = \emptyset, c$ *and* $i \neq 1$.

(ii) $\mathrm{H}^i_{\text{ét},?}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, (\frac{\cdot}{p}) \otimes \mathrm{Sym}^{\mathrm{k}}\mathrm{Kl}_2) = 0$ *for all* $? = \emptyset, c$ *and* $i \neq 1$.

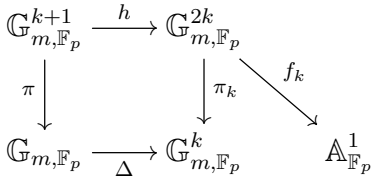
Proof. We refer the proof to [Kat88, Chapter 5]. The sheaf of Legendre symbol $\left(\frac{\bullet}{p}\right)$ is tame at both 0 and ∞; the result follows from the Grothendieck-Shafarevich-Ogg formula. □

Suppose (z, x_i) are coordinates of $\mathbb{G}^{k+1}_{m,\mathbb{F}_p}$. Let $f_k = \sum_{i=1}^k (x_i + z/x_i)$ and $\tilde{f}_k = \sum_{i=1}^k (x_i + z^2/x_i)$. Denote by $\mathscr{L}_f$ the pullback of the Artin-Schreier sheaf $f^*\,\mathscr{L}_\psi$. We have the commutative diagram:

(38)


Proposition A.1.2. *The sheaf* $\mathrm{Kl}_2^{\otimes k}$ *is isomorphic to* $\mathrm{R}^k\pi_!\mathscr{L}_{f_k}$.

Proof. Consider the following Cartesian diagram:

(39)


where

$$h(z,x_i)=(z,x_1,\dots,z,x_k) \qquad \pi(z,x_i)=z$$

$$\pi_k(y_1,x_1,\dots,y_k,x_k)=(y_1,\dots,y_k) \qquad f_k(y_1,x_1,\dots,y_k,x_k)=\sum_{i=1}^{k}(x_i+y_i/x_i)$$

and Δ is the diagonal embedding.

Note that $\mathrm{Kl}_2^{\otimes k}\simeq\Delta^*(\mathrm{Kl}_2^{\boxtimes k})$. The proposition follows from applying the spectral sequence and the base change theorem. □

Let $\chi\colon \mathfrak{S}_k,\mathfrak{S}_k\times\mu_2\to\{\pm1\}$ be the sign character of permutation group $\mathfrak{S}_k$ and $\chi'\colon \mathfrak{S}_k\times\mu_2\to\{\pm1\}$ be the sign character of permutation group and nontrivial action on μ_2. For a representation V of the finite group $G=\mathfrak{S}_k$ or $\mathfrak{S}_k\times\mu_2$ over a field K of characteristic zero, we denote $V^{G,\chi}$ or $V^{G,\chi'}$ as their isotypic parts, i.e., the image of

$$\left(\frac{1}{|G|}\sum_{\sigma\in G}\chi(\sigma)\sigma\right) \quad \text{or} \quad \left(\frac{1}{|G|}\sum_{\sigma\in G}\chi'(\sigma)\sigma\right)$$

in the group ring $K[G]$ acting on V.

Proposition A.1.3. *There are following isomorphisms.*

(i) $\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathrm{Sym}^k\mathrm{Kl}_2)\simeq\mathrm{H}^{k+1}_{\text{ét,c}}(\mathbb{G}^{k+1}_{m,\overline{\mathbb{F}}_p},\mathscr{L}_{f_k})^{\mathfrak{S}_k,\chi}\simeq\mathrm{H}^{k+1}_{\text{ét,c}}(\mathbb{G}^{k+1}_{m,\overline{\mathbb{F}}_p},\mathscr{L}_{\tilde{f}_k})^{\mathfrak{S}_k\times\mu_2,\chi}$.

(ii) $\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},(\frac{\cdot}{p})\otimes\mathrm{Sym}^{\mathrm{k}}\mathrm{Kl}_2)\simeq\mathrm{H}^{k+1}_{\text{ét,c}}(\mathbb{G}^{k+1}_{m,\overline{\mathbb{F}}_p},\mathscr{L}_{\tilde{f}_k})^{\mathfrak{S}_k\times\mu_2,\chi'}$.

Proof. We refer the proof to [FSY22, Section 2]. For the second statement, we note that

$$\mathrm{H}^{k+1}_{\text{ét,c}}(\mathbb{G}^{k+1}_{m,\overline{\mathbb{F}}_p},\mathscr{L}_{\tilde{f}_k})\simeq\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},\mathrm{Sym}^k\mathrm{Kl}_2)\oplus\mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p},(\frac{\cdot}{p})\otimes\mathrm{Sym}^k\mathrm{Kl}_2).$$

Then the second statement follows from the first one. □

Lei Fu and Daqing Wan had studied deeply the L-functions for symmetric products of the Kloosterman sums in [FW05, FW08, FW10]. We follow the ideas of Fu and Wan in [FW05] to do the computations for twisted moments of the Kloosterman sums, especially the local monodromy at ∞.

Recall that the Kloosterman sheaf is a lisse etale sheaf of rank two on $\mathbb{G}_{m,\mathbb{F}_p}$ with trace function $a \mapsto -\mathrm{Kl}_2(q;a)$ if $a \in \mathbb{G}_m(\mathbb{F}_q)$. Moreover, Kl_2 is tame at 0 and totally wild at ∞. Lisse etale sheaves on $\mathbb{G}_{m,\mathbb{F}_p}$ correspond to the Galois representations of the function field $\mathbb{F}_p(t)$ of $\mathbb{G}_{m,\mathbb{F}_p}$. By abuse of notations, we still denote the Galois representations corresponding to $\mathrm{Kl}_2, \mathrm{Sym}^k \mathrm{Kl}_2, (\frac{\cdot}{p}) \otimes \mathrm{Sym^k Kl}_2$ by $\mathrm{Kl}_2, \mathrm{Sym}^k \mathrm{Kl}_2, (\frac{\cdot}{p}) \otimes \mathrm{Sym^k Kl}_2$. Besides, we denote the decomposition group, inertia group and wild inertia group at prime $\mathfrak{p}$ by $D_\mathfrak{p}, I_\mathfrak{p}, P_\mathfrak{p}$, respectively.

First, we summarise the results of Fu-Wan in [FW05]:

- Dimension of the I_0 invariant part of $\mathrm{Sym}^k \mathrm{Kl}_2$ is 1.
- Let F_0 be the geometric Frobenius at 0. Then $\det(1 - F_0 T \mid (\mathrm{Sym}^k \mathrm{Kl}_2)^{I_0}) = 1 - T$.
- Dimension of the I_∞ invariant part of $\mathrm{Sym}^k \mathrm{Kl}_2$ is given as follow:

$$\dim(\mathrm{Sym}^k \mathrm{Kl}_2)^{I_\infty} = \begin{cases} 0 & \text{if } k \text{ is odd,} \\ 1 + [\frac{k}{2p}] & \text{if } k \equiv 0 \bmod 4, \\ [\frac{k}{2p}] & \text{if } k \equiv 2 \bmod 4. \end{cases}$$

- Let F_∞ be the geometric Frobenius at ∞. Then

$$\det(1 - F_\infty T \mid (\mathrm{Sym}^k \mathrm{Kl}_2)^{I_\infty}) = \begin{cases} 1 & \text{if } 2 \nmid k \\ (1 - p^{\frac{k}{2}} T)^{m_k} & \text{if } 2|k \text{ and } p \equiv 1 \bmod 4 \\ (1 + p^{\frac{k}{2}} T)^{n_k} (1 - p^{\frac{k}{2}} T)^{m_k - n_k} & \text{if } 2|k \text{ and } p \equiv -1 \bmod 4 \end{cases}$$

where

$$m_k = \begin{cases} 1 + [\frac{k}{2p}] & \text{if } k \equiv 0 \bmod 4 \\ [\frac{k}{2p}] & \text{if } k \equiv 2 \bmod 4 \end{cases}$$

and $n_k = [\frac{k}{4p} + \frac{1}{2}]$.

Now, we start to prove the analogous results for twisted moments of the Kloosterman sums.

Proposition A.1.4. *For I_0 invariant part, we have* $\dim((\frac{\cdot}{p}) \otimes \mathrm{Sym^k Kl}_2)^{I_0} = 0$.

Proof. By [Del77, Kat88], the monodromy of Kl_2 at 0 is unipotent with a single Jordan block. Let γ be a topological generator of the tame quotient of I_0. Then there exists a basis $\{v_1, v_2\}$ of Kl_2 such that $\gamma v_1 = v_1$ and $\gamma v_2 = v_1 + v_2$. Let $w_i = v_1^{k-i} v_2^i \in \mathrm{Sym}^k \mathrm{Kl}_2$. Then $\{w_i\}_{i=0}^k$ forms a basis of $\mathrm{Sym}^k \mathrm{Kl}_2$.

Suppose $w = \sum_{i=0}^k a_i w_i \in ((\frac{\cdot}{p}) \otimes \mathrm{Sym}^k \mathrm{Kl}_2)^{I_0}$. Recall that $(\frac{\cdot}{p})$ is tamely ramified at 0 with $\gamma|_{(\frac{\cdot}{p})} = -1$. Therefore, we have

$$-a_0 v_1^k - a_1 v_1^{k-1}(v_1 + v_2) - \cdots - a_k (v_1 + v_2)^k = a_0 v_1^k + a_1 v_1^{k-1} v_2 + \cdots + a_k v_2^k.$$

By comparing the coefficients of $v_2^k, v_1 v_2^{k-1}, \cdots, v_1^k$ successively, we deduce that all a_i vanish. □

Corollary A.1.5. *Let F_0 be the geometric Frobenius at 0. Then*

$$\det(1 - F_0 T | ((\frac{\cdot}{p}) \otimes \mathrm{Sym^k Kl}_2)^{I_0}) = 1.$$

Proof. This follows straightforwardly from Proposition A.1.4. □

Next, we investigate the local behaviour of the representations at ∞. We will use the same notations as Fu and Wan, and we state those terminologies first:

Let p be an odd prime. Define

$$S_k(2,p) = \{(j_0, j_1) \in \mathbb{Z}^2_{\geq 0} \mid j_0 - j_1 \equiv 0 \bmod p \text{ and } j_0 + j_1 = k\}.$$

Let σ be the switching operator acting on $S_k(2,p)$ by $\sigma(j_0, j_1) = (j_1, j_0)$.

Suppose V is a vector space with basis $\{e_0, e_1\}$. Denote

$$v_j = e_0^j e_1^{k-j} + (-1)^{k-j} e_0^{k-j} e_1^j$$

as elements in $\mathrm{Sym}^k V$.

When k is even, let $b_k(2,p)$ be the number of σ-orbits in $S_k(2,p)$ such that the subspace spanned by that orbit is not zero. More explicitly,

$$b_k(2,p) = \begin{cases} 1 + [\frac{k}{2p}] & \text{if } k \equiv 0 \bmod 4, \\ [\frac{k}{2p}] & \text{if } k \equiv 2 \bmod 4. \end{cases}$$

Proposition A.1.6. *Dimension of the I_∞ invariant part of $(\frac{\cdot}{p}) \otimes \mathrm{Sym^k Kl_2}$ is given by the following formula:*

$$\dim((\frac{\cdot}{p}) \otimes \mathrm{Sym^k Kl_2})^{I_\infty} = \begin{cases} 0 & \textit{if } k \textit{ is odd}, \\ [\frac{k}{2p}] & \textit{if } k \equiv 0 \bmod 4, \\ 1 + [\frac{k}{2p}] & \textit{if } k \equiv 2 \bmod 4. \end{cases}$$

Proof. We use the same notations as Fu-Wan in [FW05, Lemma 2.1]. There exists a basis $\{e_0, e_1\}$ of Kl_2 such that $ge_0 = e_1$ and $ge_1 = -e_0$. Besides, we also have $g_{(a,\mu)}(e_i) = \psi_2((-1)^{i+1}a)\chi(\mu^{-1})e_i$ where $(a, \mu) \in \mathbb{F}_p \times \mu_2$.

Suppose $v = \sum_{j=0}^k a_j e_0^j e_1^{k-j}$ is invariant under the actions of I_∞. Then we have equation $gv = v$, that is:

$$-(\sum_{j=0}^k (-1)^{k-j} a_j e_1^j e_0^{k-j}) = \sum_{j=0}^k a_j e_0^j e_1^{k-j}.$$

(The minus sign on the left hand side is due to the twist of the Legendre symbol.) By comparing the coefficients of both sides, we deduce that

$$\begin{cases} a_j = (-1)^{j+1} a_{k-j}, \\ a_{k-j} = (-1)^{k-j+1} a_j. \end{cases}$$

Therefore, we have $a_j = (-1)^{k+2} a_j$. If k is an odd number, then $a_j = 0$ for all j and hence $v = 0$. If k is an even number, then v is generated by $\tilde{v}_j = e_0^j e_1^{k-j} + (-1)^{k-j+1} e_0^{k-j} e_1^j$ with $0 \leq j \leq \frac{k}{2}$.

Moreover, we also have equations $g_{(a,1)}(v) = v$ for all $a \in \mathbb{F}_p$, which are:

$$\sum_{j=0}^k a_j \psi_2(-ja + (k-j)a) e_0^j e_1^{k-j} = \sum_{j=0}^k a_j e_0^j e_1^{k-j}.$$

Therefore, we deduce that $j-(k-j)\equiv 0 \bmod p$. Hence, we conclude that

$$\dim((\frac{\cdot}{p})\otimes \mathrm{Sym^k Kl_2})^{I_\infty} = \begin{cases} 0 & \text{if } k \text{ is odd,} \\ b_k(2,p) & \text{if } k \text{ is even.} \end{cases}$$

Finally, we observe that $\tilde{v}_{\frac{k}{2}}$ is zero if and only if $4|k$. This completes the proof. □

Proposition A.1.7. *Let F_∞ be the geometric Frobenius at ∞. Then*

$$\det(1-F_\infty T \mid ((\frac{\cdot}{p})\otimes \mathrm{Sym}^k \mathrm{Kl}_2)^{I_\infty}) = \begin{cases} 1 & \text{if } 2\nmid k \\ (1-p^{\frac{k}{2}}T)^{m_k} & \text{if } 2|k \text{ and } p\equiv 1 \quad (4) \\ (1+p^{\frac{k}{2}}T)^{n_k}(1-p^{\frac{k}{2}}T)^{m_k-n_k} & \text{if } 2|k \text{ and } p\equiv -1 \quad (4) \end{cases}$$

where

$$m_k = \begin{cases} [\frac{k}{2p}] & \text{if } k\equiv 0 \bmod 4 \\ 1+[\frac{k}{2p}] & \text{if } k\equiv 2 \bmod 4 \end{cases}$$

and $n_k = [\frac{k}{4p}+\frac{1}{2}]$.

Proof. This is an analog of the result of Fu-Wan in [FW05, Lemma 4.1]. The proof only requires a similar modification as that of Proposition A.1.6. □

Theorem A.1.8. *For any odd prime $p>k$,*

$$\dim(\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, (\frac{\cdot}{p})\otimes \mathrm{Sym^k Kl_2})) = \begin{cases} \frac{k+1}{2} & \text{if } k \text{ is odd,} \\ \frac{k}{2} & \text{if } k\equiv 0 \bmod 4, \\ \frac{k}{2}-1 & \text{if } k\equiv 2 \bmod 4. \end{cases}$$

Proof. For $p>k$, we may compute the dimension by

$$\begin{aligned} \dim \mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, (\frac{\cdot}{p})\otimes \mathrm{Sym}^k \mathrm{Kl}_2) &= \dim \mathrm{H}^1_{\text{ét,c}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, (\frac{\cdot}{p})\otimes \mathrm{Sym}^k \mathrm{Kl}_2) \\ &\quad - \dim((\frac{\cdot}{p})\otimes \mathrm{Sym}^k \mathrm{Kl}_2)^{I_0} - \dim((\frac{\cdot}{p})\otimes \mathrm{Sym}^k \mathrm{Kl}_2)^{I_\infty}. \end{aligned}$$

Hence, we have

$$\dim(\mathrm{H}^1_{\text{ét,mid}}(\mathbb{G}_{\mathrm{m},\overline{\mathbb{F}}_p}, (\frac{\cdot}{p})\otimes \mathrm{Sym}^k \mathrm{Kl}_2)) = \begin{cases} \frac{k+1}{2}-0-0 & \text{if } k \text{ is odd,} \\ \frac{k}{2}-0-[\frac{k}{2p}] & \text{if } k\equiv 0 \bmod 4, \\ \frac{k}{2}-0-(1+[\frac{k}{2p}]) & \text{if } k\equiv 2 \bmod 4. \end{cases}$$

Note that $[\frac{k}{2p}]=0$ when $p>k$, and this completes the proof. □

A.2. **Kloosterman sums and hypergeometric character sums.** Over the past few decades, progress has been made in understanding the relationship between Kloosterman sums and hypergeometric functions over finite fields, see [Kat88, Gre87, Eva10, LT15, KAS25, Sai25, Ots24] for example. The modularity of the Kloosterman representations can also be obtained from the modularity of the corresponding hypergeometric representations. In this subsection, we summarize some of the identities related to Evans's conjectures.

For simplicity, we fix $\psi\colon \mathbb{F}_q \to \mathbb{C}^\times$ to be the canonical additive character of $\mathbb{F}_q$ and extend the multiplicative characters on $\mathbb{F}_q^\times$ to $\mathbb{F}_q$ by setting $\chi(0) = 0$. Denote the $g(A)$ and $J(A, B)$ the Gauss sum and Jacobi sum:

$$g(A) := \sum_{a\in\mathbb{F}_q} A(a)\psi(a), \quad J(A, B) = \sum_{a\in\mathbb{F}_q} A(a)B(1 - a),$$

where A, $B \in \widehat{\mathbb{F}_q^\times}$.

Built on the orthogonality relations and the fact that

$$\psi(a) = \frac{1}{q-1}\sum_{\chi\in\widehat{\mathbb{F}_q^\times}} g(\overline{\chi})\chi(a), \quad a \in \mathbb{F}_q^\times,$$

we can write $\mathsf{m}_k^+(q)$ and $\mathsf{m}_k(q)$ in terms of Gauss and Jacobi sums.

Lemma A.2.1. *For $a \in \mathbb{F}_q^\times$,*

$$\mathrm{Kl}_2(q; a) = \frac{1}{q-1}\sum_{\chi\in\widehat{\mathbb{F}_q^\times}} g(\overline{\chi})^2\chi(a),$$

$$\mathrm{Kl}_2^{\mathrm{Sym}^2}(a; q) = \mathrm{Kl}_2(q; a)^2 - q = \frac{1}{q-1}\frac{1}{g(\phi)}\sum_{\chi\in\widehat{\mathbb{F}_q^\times}} g(\overline{\chi})^3 g(\phi\chi)\chi(-4a),$$

where ϕ is the quadratic character of $\widehat{\mathbb{F}_q^\times}$.

Corollary A.2.2. *For an odd prime power q, let $\phi \in \widehat{\mathbb{F}_q^\times}$ be the quadratic character. Then*

$$\mathsf{m}_1(q) = -g(\phi)^2 = -\phi(-1)q, \qquad \mathsf{m}_2(q) = -\phi(-1)g(\phi)^2 = -q.$$

Further, we define hypergeometric functions over finite fields as follows. Let A be a multiplicative character and $t \in \mathbb{F}_q^\times$, we define

$${}_1\mathbb{P}_0[A; t; q] := \overline{A}(1 - t).$$

When $n \geq 2$, for any A_i, $B_j \in \widehat{\mathbb{F}_q^\times}$, $i = 0, \ldots, n-1$, $j = 1, \ldots, n-1$, we define

$$\begin{aligned}
{}_n\mathbb{P}_{n-1}\left[\begin{matrix} A_0 & A_1 & \cdots & A_{n-1} \\ & B_1 & \cdots & B_{n-1}\end{matrix}; t; q\right] :=& \\
\sum_{x\in\mathbb{F}_q} A_{n-1}(x)\overline{A}_{n-1}B_{n-1}(1-x)\cdot {}_{n-1}\mathbb{P}_{n-2}&\left[\begin{matrix} A_0 & A_1 & \cdots & A_{n-2} \\ & B_1 & \cdots & B_{n-2}\end{matrix}; tx; q\right] \\
= \frac{(-1)^n}{q-1}\left(\prod_{i=1}^{n-1} A_iB_i(-1)\right)&\sum_{\chi\in\widehat{\mathbb{F}_q^\times}} \binom{A_0\chi}{\chi}\binom{A_1\chi}{B_1\chi}\cdots\binom{A_{n-1}\chi}{B_{n-1}\chi}\chi(t),
\end{aligned}$$

where $\binom{A}{B} := -B(-1)J(A, \overline{B})$, for any characters A, B. There are different versions of hypergeometric type character sums documented in literature. Here we use the version given in [FLR+22], for the purpose that these are closely related to the hypergeometric sheaves introduced by Katz [Kat88, Kat96, LLT22].

From the inductive formula

$$\mathrm{Kl}_2^{\mathrm{Sym}^n}(a;q) = -\mathrm{Kl}_2(a;q)\,\mathrm{Kl}_2^{\mathrm{Sym}^{n-1}}(a;q) - q\,\mathrm{Kl}_2^{\mathrm{Sym}^{n-2}}(a;q) \tag{40}$$

and Lemma A.2.1, one can induce the third and fourth moments as $\mathbb{P}$-values (see [LT15, KAS25, Sai25] for example), which we list the relation below. We will drop the q-notation (in our case $q = p$) in the $\mathbb{P}$-function.

Lemma A.2.3. *For $a \in \mathbb{F}_p^\times$,*

$$-\sum_{a\in\mathbb{F}_p^\times} \phi(a)\,\mathrm{Kl}_2(a;p)\,\mathrm{Kl}_2^{\mathrm{Sym}^2(a;p)} = p\phi(-1)\,{}_3\mathbb{P}_2\left[\begin{matrix}\phi & \phi & \phi\\ & \varepsilon & \varepsilon\end{matrix};\ 4\right] = p\,{}_3\mathbb{P}_2\left[\begin{matrix}\phi & \phi & \phi\\ & \varepsilon & \varepsilon\end{matrix};\ \frac{1}{4}\right],$$

$$\sum_{a\in\mathbb{F}_p^\times} \phi(a)\left(\mathrm{Kl}_2^{\mathrm{Sym}^2(a;p)}\right)^2 = p\phi(-1)\,{}_4\mathbb{P}_3\left[\begin{matrix}\phi & \phi & \phi & \phi\\ & \varepsilon & \varepsilon & \varepsilon\end{matrix};\ 1\right].$$

where ϕ is the quadratic character of $\widehat{\mathbb{F}_p^\times}$.

Theorem A.2.4. [AO00, AOP02, Eva10] *Let $p > 3$ be an odd prime, and ϕ, ε be the quadratic character and trivial character of $\mathbb{F}_p^\times$, respectively. Then*

$$\mathsf{m}_3(p) = -\,p\,{}_3\mathbb{P}_2\left[\begin{matrix}\phi & \phi & \phi\\ & \varepsilon & \varepsilon\end{matrix};\ \frac{1}{4}\right] + \phi(-1)p^2 = -p\cdot a_p(f_{48.3.e.a}),$$

$$\mathsf{m}_4(p) = \phi(-1)p\cdot\,{}_4\mathbb{P}_3\left[\begin{matrix}\phi & \phi & \phi & \phi\\ & \varepsilon & \varepsilon & \varepsilon\end{matrix};\ 1\right] + p^2 = -p\cdot a_p(f_{8.4.a.a}),$$

where $a_p(f)$ is the pth Fourier coefficient of the Hecke eigenform f.

For higher powers k, the expressions involve inductive formula of $\mathbb{P}$-functions.

Proposition A.2.5. *Let $p > 3$ be an odd prime, and ϕ, ε be quadratic character and trivial character of $\mathbb{F}_p^\times$, respectively. We have*

$$\mathsf{m}_5(p) = -\,p\sum_{\substack{t\\ y\neq 0,1,1/4t}} \phi\left(\frac{t}{t-1}\right)\,{}_3\mathbb{P}_2\left[\begin{matrix}\phi & \phi & \phi\\ & \varepsilon & \varepsilon\end{matrix};\ (1-y)(t-\frac{1}{4y})\right]$$
$$+\,2p^2\,{}_3\mathbb{P}_2\left[\begin{matrix}\phi & \phi & \phi\\ & \varepsilon & \varepsilon\end{matrix};\ \frac{1}{4}\right] + 2p + p(p+1)\left(\frac{-3}{p}\right).$$

To express m_5 in terms of $\mathbb{P}$-functions, we will apply the following basic properties.

Lemma A.2.6. *Let A, $B \in \widehat{\mathbb{F}_q^\times}$.*

$$J(A,B) = \frac{g(A)g(B)}{g(AB)}, \quad AB \neq \varepsilon,$$
$$g(A)g(\overline{A}) = A(-1)q, \quad A \neq \varepsilon,$$

and, for $a \in \mathbb{F}_q^\times$,

$$\frac{1}{q-1}\sum_{K\in\widehat{\mathbb{F}_q^\times}} J(AK,\overline{K})K(a) = A(1-a).$$

Proof. From the inductive formula (40), for a fixed prime p, we can simplify the term $\mathrm{Kl}_2^{\mathrm{Sym}^5}(a;p)$ as

$$\begin{aligned}\mathrm{Kl}_2^{\mathrm{Sym}^5}(a;p) =& \mathrm{Kl}_2^{\mathrm{Sym}^1}(a;p)\,\mathrm{Kl}_2^{\mathrm{Sym}^4(a;p)} - p\cdot \mathrm{Kl}_2^{\mathrm{Sym}^3}(a;p)\\ =& \mathrm{Kl}_2^{\mathrm{Sym}^1}(a;p)\left(\mathrm{Kl}_2^{\mathrm{Sym}^2}(a;p)\right)^2 - 2p\,\mathrm{Kl}_2^{\mathrm{Sym}^1}(a;p)\,\mathrm{Kl}_2^{\mathrm{Sym}^2}(a;p),\end{aligned}$$

and hence,

$$\mathsf{m}_5(p) = \sum_{a\in\mathbb{F}_q}\phi(a)\,\mathrm{Kl}_2^{\mathrm{Sym}^1}(a;p)\left(\mathrm{Kl}_2^{\mathrm{Sym}^2}(a;p)\right)^2 - 2p\sum_{a\in\mathbb{F}_q}\phi(a)\,\mathrm{Kl}_2^{\mathrm{Sym}^1}(a;p)\,\mathrm{Kl}_2^{\mathrm{Sym}^2}(a;p).$$

For short hand, we write $\mathsf{m}_5(p) = I_1 + I_2$, where I_1 and I_2 are the first and second terms in the above identity, respectively. Note the I_2 is related to $\mathsf{m}_3(p)$ and is

$$-I_2 = 2p^2\,{}_3\mathbb{P}_2\left[\begin{matrix}\phi & \phi & \phi\\ & \varepsilon & \varepsilon\end{matrix}; \frac{1}{4}\right],$$

which we will omit the detail here.

For I_1, by Lemma A.2.1 and orthogonality property, we deduce that

$$\begin{aligned}-I_1 =& \frac{1}{p(p-1)^2}\sum_{\chi\in\widehat{\mathbb{F}_p^\times}} g(\overline{\chi})^3 g(\phi\chi)\sum_{K\in\widehat{\mathbb{F}_p^\times}} g(\overline{K})^2 g(\overline{K\chi})g(\phi\chi K)^3\overline{K}(-4)\\ \overset{\chi\mapsto\phi\overline{\chi}}{=}& \frac{1}{p(p-1)^2}\sum_{\chi\in\widehat{\mathbb{F}_p^\times}} g(\phi\chi)^3 g(\overline{\chi})\sum_{K\in\widehat{\mathbb{F}_p^\times}} g(\overline{K})^2 g(\phi\chi\overline{K})g(\overline{\chi}K)^3\overline{K}(-4).\end{aligned}$$

We now apply the basic properties in Lemma A.2.6 to I_1. Firstly, we split I_1 into two parts:

$$\begin{aligned}I_1 =& \frac{g(\phi)}{p(p-1)^2}\sum_{\chi\neq\varepsilon} g(\phi\chi)^3 g(\overline{\chi})^3\sum_{K\in\widehat{\mathbb{F}_p^\times}} J(\overline{\chi}K,\overline{K})^2 J(\overline{\chi}K,\phi\chi\overline{K})\overline{K}(-4)\\ &- \frac{g(\phi)^3}{p(p-1)^2}\sum_{K\in\widehat{\mathbb{F}_p^\times}} g(\overline{K})^2 g(\phi\overline{K})g(K)^3\overline{K}(-4)\\ =& I_{1,1} + I_{1,2},\end{aligned}$$

where

$$\begin{aligned}I_{1,1} :=& \frac{g(\phi)}{p(p-1)^2}\sum_{\chi\in\widehat{\mathbb{F}_p^\times}} g(\phi\chi)^3 g(\overline{\chi})^3\sum_{K\in\widehat{\mathbb{F}_p^\times}} J(\overline{\chi}\overline{K},K)^2 J(\overline{\chi}\overline{K},\phi\chi K)K(-4),\\ I_{1,2} :=& \frac{g(\phi)^4}{p(p-1)^2}\sum_{K\in\widehat{\mathbb{F}_p^\times}} J(K,\overline{K})^2 J(\overline{K},\phi K)\overline{K}(-4)\\ &- \frac{g(\phi)^3}{p(p-1)^2}\sum_{K\in\widehat{\mathbb{F}_p^\times}} g(\overline{K})^2 g(\phi\overline{K})g(K)^3 K(-4).\end{aligned}$$

By Lemma A.2.6, we can further simplify $I_{1,2}$ as

$$\begin{aligned}
I_{1,2} =& \frac{g(\phi)^4}{p(p-1)^2}\left(\sum_{K\in\widehat{\mathbb{F}_p^\times}} J(\overline{K},\phi K)\overline{K}(-4) - J(\varepsilon,\phi) - J(\varepsilon,\varepsilon)^2\right) \\
& - \frac{g(\phi)^4}{p(p-1)^2}\left(p^2\cdot\sum_{K\in\widehat{\mathbb{F}_p^\times}} J(K,\phi\overline{K})K(-4) - p^2\cdot J(\varepsilon,\phi) - 1\right) \\
=& \frac{p^2}{p(p-1)^2}\left(\phi(-3)(p-1) + 1 - (p-2)^2 - \phi(-3)p^2(p-1) - p^2 + 1\right) \\
=& -\phi(-3)p(p+1) - 2p.
\end{aligned}$$

Regarding $I_{1,2}$, we will make a change of variable using $K\mapsto\overline{\chi}K$ to make the expression simpler. That is,

$$\begin{aligned}
I_{1,1} =& \frac{g(\phi)}{p(p-1)^2}\sum_{\chi\in\widehat{\mathbb{F}_p^\times}} g(\phi\chi)^3 g(\overline{\chi})^3\chi(-4)\sum_{K\in\widehat{\mathbb{F}_p^\times}} J(\overline{\chi}K,\overline{K})^2 J(\phi K,\overline{K})K(-4), \\
=& -\frac{g(\phi)^4}{p(p-1)^2}\sum_{\chi\in\widehat{\mathbb{F}_p^\times}} J(\phi\chi,\overline{\chi})^3\overline{\chi}(4)\cdot(p-1)(-\phi(-1))\,{}_3\mathbb{P}_2\left[\begin{matrix}\overline{\chi} & \overline{\chi} & \phi \\ & \varepsilon & \varepsilon\end{matrix};\,4\right] \\
=& \frac{\phi(-1)p}{p-1}\sum_{\chi\in\widehat{\mathbb{F}_p^\times}}\binom{\phi\chi}{\chi}^3\chi(-1/4)\cdot\,{}_3\mathbb{P}_2\left[\begin{matrix}\overline{\chi} & \overline{\chi} & \phi \\ & \varepsilon & \varepsilon\end{matrix};\,4\right].
\end{aligned}$$

From the $\mathbb{P}$-expression, we can write

$${}_3\mathbb{P}_2\left[\begin{matrix}\overline{\chi} & \overline{\chi} & \phi \\ & \varepsilon & \varepsilon\end{matrix};\,4\right] = \sum_{t,y}\phi\left(\frac{t}{1-t}\right)\overline{\chi}(y)\chi(1-y)\chi(1-4ty).$$

This gives

$$\begin{aligned}
I_{1,1} &= \frac{\phi(-1)p}{p-1}\sum_{t,y}\phi\left(\frac{t}{1-t}\right)\sum_{\chi\in\widehat{\mathbb{F}_p^\times}}\binom{\phi\chi}{\chi}^3\chi\left(\frac{-(1-y)(1-4ty)}{4y}\right) \\
&= \phi(-1)\sum_{t,y}\phi\left(\frac{t}{1-t}\right)\,{}_3\mathbb{P}_2\left[\begin{matrix}\phi & \phi & \phi \\ & \varepsilon & \varepsilon\end{matrix};\,(1-y)(t-\frac{1}{4y})\right],
\end{aligned}$$

which leads to the desired expression in the Proposition. □

Remark A.2.7. As an application in the theory of modular forms, it is worthy to mention that it would be interesting to investigate relation between the double sums involved hypergeometric functions in the proposition and certain space of cusp forms. In the classical case, the space of cusp forms on $\Gamma_1(N)$ are spanned by Poincare series, whose Fourier coefficients are (twisted) Kloosterman sums (see [CS17, Section 9.3] for example). In the works of [Sch88, HLLT24, FOP04, Fus10, Len11b, Len11a], the traces of Hecke operators on the certain subgroups of triangle groups commensurable with $\mathrm{PSL}_2(\mathbb{Z})$ can be computed via

$\mathbb{P}$-functions and suitable rational functions as the arguments. It might be interesting to see the connection from what the modularity of the Kloosterman sheaves suggest.

A.3. **L-functions.** We list what we know about the shapes of the L-functions $L_k(s)$ and the completed one $\Lambda_k(s)$ associated with the Galois representations $\rho_{k,\ell}$ (or the motive M_k) in Theorem 1.2.1, as in [FSY22, §5] and [FSY23, §8]. Since $\rho_{k,\ell}$ is potentially automorphic, $\Lambda_k(s)$ extends meromorphically to $\mathbb{C}$ and admits the functional equation $\Lambda_k(k+2-s) = \pm\Lambda_k(s)$.

Euler factors. By Propositions 3.1.2 and 3.1.3, the local Euler factor $L_k(p;p^{-s})^{-1}$ of $L_k(s)$ at an odd prime p is given by

$$\begin{aligned} L_k(p;T) &= \det(1 - F_p T \mid V_{k,\ell}^{I_p}) \\ &= \begin{cases} Z_k(p;T) \prod_{a \in \Theta_k(p)^+} \Big(1 - \Big(\frac{(-1)^{(1+ap)/2} 2a'}{p}\Big) p^{(k+1)/2} T\Big) & \text{for } k \text{ odd}, \\ Z_k(p;T)(1-p^{k/2})^{-\delta_{2+4\mathbb{Z}}(k)} & \text{for } k \text{ even}, \end{cases} \end{aligned}$$

where $Z_k(p;T)$ and $\Theta_k(p)^+$ are defined in (17) and (28), respectively, and $a' = ap^{-v_p(a)}$ is the prime-to-p part of a. For $k \le 6$, the local Euler factor at 2 is trivial.

Conductors. In general, the compatible system $\{\rho_{k,\ell}\}_\ell$ is wildly ramified at prime 2. We write its conductor as

$$\mathfrak{N}_k = 2^c \mathfrak{N}_k'$$

where $\mathfrak{N}_k'$ is odd. We have

$$\mathfrak{N}_k' = \begin{cases} 1_{\boxtimes} 3_{\boxtimes} 5_{\boxtimes} \cdots k_{\boxtimes}, & k \text{ odd}, \\ 2_{\triangle} 4_{\triangle} 6_{\triangle} \cdots k_{\triangle}, & k \text{ even}. \end{cases}$$

Here $n_{\boxtimes}$ (the square-free part) is the product of prime divisors p of n with odd valuation $v_p(n)$, and $n_{\triangle}$ (the odd part of the radical) is the product of odd prime divisors. For example, for $k = 1, 3, 5$, we have $c = 2, 4, 6$, respectively, while for $k = 4, 6$, $c = 3$.

Gamma factors. The gamma factor of M_k is determined by the Hodge numbers of its de Rham realization $V_{k,\mathrm{dR}}$ together with the action ι induced by the complex conjugation. Notice that ι switches $V_{k,\mathrm{dR}}^{pq}$ and $V_{k,\mathrm{dR}}^{qp}$ in the Hodge decomposition, and $V_{k,\mathrm{dR}}^{(k+1)/2,(k+1)/2} \neq 0$ only when $k \equiv 1 \bmod 4$. In the latter, $h^{(k+1)/2,(k+1)/2} = 1$. Consequently, by (29), ι acts on $V_{k,\mathrm{dR}}^{(k+1)/2,(k+1)/2}$ by

$$(-1)^{(k-1)/4} \det \rho_{k,\ell}(\iota) = 1 = -(-1)^{(k+1)/2}.$$

The gamma factor of M_k is then given by

$$L_k(\infty, s) = \pi^{-ms/2} \prod_{j=1}^{m} \Gamma\Big(\frac{s+1-j}{2}\Big), \quad m = \operatorname{rk} \mathsf{M}_k = \Big\lfloor \frac{k+1}{2} \Big\rfloor - \delta_{2+4\mathbb{Z}}(k).$$

Periods. For the calculation of the period realizations of the twisted moments of the Bessel connection Kl_2 over $\mathbb{Q}$, see [Chu24]. Following the strategy of [FSY23, §7], we now show that they coincide with the period realizations of the motives M_k, focusing on constructing certain rapid decay cycle classes for $(U_0, \tilde{f}_k)$ (i.e., rapid decay for the function $e^{-\tilde{f}_k}$). Here, $U_0 = \mathbb{G}_{\mathrm{m}}^{k+1}$ over $\mathbb{Q}$ with Cartesian coordinates $\{t, y_i\}_{i=1}^k$ and $\tilde{f}_k = \frac{t}{2}\sum_{i=1}^k y_i + y_i^{-1}$, regarded as a regular function on U_0.[1] Consider the action of $\mathfrak{S}_k \times \mu_2$ on $(U_0, \tilde{f}_k)$, extending that on $\mathscr{K} \subset \mathbb{G}_{\mathrm{m}}^k$ in the beginning of §3.1, where $\mathfrak{S}_k$ acts trivially on the variable t and the generator of μ_2 sends t to $-t$. One has $\mathsf{M}_k = W_{k+1}\mathrm{H}^{k+1}(U_0, \tilde{f}_k)_\chi$, the χ-isotypic part of the weight $k+1$ piece.

Let $U = U_0(\mathbb{C})$. We consider (oriented, unbounded) chains on U. Let c_p be the circles $|t| = 1$ if $p = 0$, and $|y_p| = 1$ for $1 \le p \le k$, equipped with the counterclockwise orientation. Consider the chains

$$\tilde{\alpha}_i' = [1, \infty) \times \big(\textstyle\prod_{p=1}^i c_p\big) \times \mathbb{R}_{>0}^{k-i}, \quad 0 \le i \le \lfloor (k-1)/2 \rfloor$$

with boundaries

$$\partial\tilde{\alpha}_i' = -\{1\} \times \big(\textstyle\prod_{p=1}^i c_p\big) \times \mathbb{R}_{>0}^{k-i}.$$

For $1 \le r \le k$, define the chain ([FSY23]) $\tilde{\gamma}_r \colon [0,1]^r \times (0,1)^{k+1-r} \to U$ by

$$(s_0, \dots, s_k) \mapsto \begin{cases} t = e^{2\pi i \cdot s_0}, & \\ y_p = e^{2\pi i \cdot s_p}, & 1 \le p < r, \\ y_p = \begin{cases} e^{2\pi i \cdot s_0} \cdot 3s_p, & 0 < s_p \le 1/3, \\ e^{2\pi i \cdot s_0(3-6s_p)}, & 1/3 \le s_p \le 2/3, \\ e^{-2\pi i \cdot s_0} \cdot \frac{1}{2}\frac{s_p}{1-s_p}, & 2/3 \le s_p < 1, \end{cases} & r \le p \le k. \end{cases}$$

In particular, $\partial\tilde{\gamma}_k = -2\{1\} \times \prod_{p=1}^k c_p$. Consider the half of $\tilde{\gamma}_r$

$$\tilde{\xi}_r(s_0, s_p) = \tilde{\gamma}_r(2^{-1}s_0, s_p), \quad \begin{matrix} 0 \le s_p \le 1 & \text{if } 0 \le p < r, \\ 0 < s_p < 1 & \text{if } r \le p \le k, \end{matrix}$$

with boundary

$$\partial\tilde{\xi}_r = \{-1\} \times (\textstyle\prod_{p=1}^{r-1} c_p) \times \prod_{p=r}^k (\mathbb{R}_{<0} - c_p) - \{1\} \times (\prod_{p=1}^{r-1} c_p) \times \mathbb{R}_{>0}^{k-r+1}.$$

Then the function $e^{-\tilde{f}_k}$ decays rapidly along the chains $\tilde{\alpha}_i', \tilde{\gamma}_r, \tilde{\xi}_r$. We glue these chains to form cycles.

Let $(\vartheta_n \in \mathbb{Q})_{n \ge 0}$ be the sequence given by $\vartheta_0 = 1$ and the recursive relations[2]

$$\sum_{p=0}^{n-1} \binom{n}{p} \vartheta_p + 2\vartheta_n = 0, \quad n \ge 1. \tag{41}$$

Set $\tilde{\xi}_{k+1} = \tilde{\gamma}_k$. For $0 \le i \le \lfloor (k-1)/2 \rfloor$, let

$$\tilde{\alpha}_i = \frac{1}{|\mathfrak{S}_{k-i} \times \mu_2|} \sum_{g \in \mathfrak{S}_{k-i} \times \mu_2} \chi(g) g\Big(\tilde{\alpha}_i' - \frac{1}{2}\sum_{r=0}^{k-i} (-1)^r \binom{k-i}{r} \vartheta_r \tilde{\xi}_{i+r+1}\Big).$$

[1] One uses the change of variables $z = (t/2)^2, x_i = (t/2)y_i$ so that $\sum_{i=1}^k x_i + z x_i^{-1} = \tilde{f}_k$.

[2] In fact, ϑ_n is the constant term of the Euler polynomial $E_n(x)$ so that $\sum_{n=0}^\infty \vartheta_n \frac{z^n}{n!} = \frac{2}{e^z+1}$ ([Chu24, (15)]).

where we regard $\mathfrak{S}_{k-i}\times\mu_2$ as a subgroup of $\mathfrak{S}_k\times\mu_2$ by letting $\mathfrak{S}_{k-i}$ act on the last $k-i$ components. By (41), one checks readily that $\partial\tilde{\alpha}_i=0$.

Proposition A.3.1. *The period structure of* M_k *coincides with the one obtained in* [Chu24, Cor.26].

Proof. Consider the χ-symmetrization of $\tilde{\alpha}_i$

$$\tau_i=\frac{1}{|\mathfrak{S}_k\times\mu_2|}\sum_{g\in\mathfrak{S}_k\times\mu_2}\chi(g)g(\tilde{\alpha}_i),\quad 0\le i\le\left\lfloor\frac{k-1}{2}\right\rfloor.$$

They define elements in the χ-isotypic part of the rapid decay homology of middle degree of $(U,\tilde{f}_k)$ with rational coefficients. Let $w_j=t^{2j}\mathrm{d}t\prod_{p=1}^{k}\mathrm{d}\log y_p, 0\le j\le\lfloor(k-1)/2\rfloor$, regarded as elements in the χ-isotypic part of the middle de Rham cohomology of $(U_0,\tilde{f}_k)$ over $\mathbb{Q}$. Direct calculation of the periods of the cycles τ_i against the forms w_j yields the assertion (cf., the proof of [FSY23, Cor.7.12(2)]). □

Deligne's Conjecture. This is about the special values of the L-functions. Similar to [FSY23, §8], one can compute explicitly the so-called critical values c_n, up to an algebraic multiple. The critical integers n and the values c_n are listed in the following Table 1. The conjecture then says that the value $L_k(n)$ is a rational multiple of c_n.

We define

$$\begin{aligned}D_{k,\mathrm{even}}&=\det\left(\int_0^\infty I_0(t)^{2i}K_0(t)^{k-2i}t^{2j}\,\mathrm{d}t\right)_{0\le i,j\le\lfloor(k-1)/4\rfloor},\\ D_{k,\mathrm{odd}}&=\det\left(\int_0^\infty I_0(t)^{2i+1}K_0(t)^{k-1-2i}t^{2j}\,\mathrm{d}t\right)_{0\le i,j\le\lfloor(k-3)/4\rfloor}.\end{aligned}\tag{42}$$

If $k\equiv 2 \bmod 4$, we let $D'_{k,\mathrm{even}}$ be the determinant of the submatrix obtained by removing the last rows and columns of the matrix in (42).

For example, direct computations yield

$$D_{1,\mathrm{even}}=\int_0^\infty K_0(t)\,\mathrm{d}t=\frac{\pi}{2},\quad D_{1,\mathrm{odd}}=1,\quad D_{2,\mathrm{even}}=\int_0^\infty K_0(t)^2\,\mathrm{d}t=\frac{\pi^2}{4}.$$

Furthermore, let $\chi=\left(\frac{-1}{\cdot}\right), g=f_{60.3.b.a}$ as appeared in Theorem 1.2.2(iii). We have[3]

$$\begin{aligned}D_{3,\mathrm{even}}&=\int_0^\infty K_0(t)^3\,\mathrm{d}t=\frac{\sqrt{3}\pi^3}{8}\,{}_3F_2\left[\begin{matrix}\frac12&\frac12&\frac12\\&1&1\end{matrix};\frac14\right]=\frac{\pi^2}{2}L(f_{48.3.e.a},1),\\ D_{3,\mathrm{odd}}&=\int_0^\infty I_0(t)K_0(t)^2\,\mathrm{d}t=\frac{\pi^2}{4}\,{}_3F_2\left[\begin{matrix}\frac12&\frac12&\frac12\\&1&1\end{matrix};\frac14\right]=2L(f_{48.3.e.a},2),\\ D_{4,\mathrm{even}}&=\int_0^\infty K_0(t)^4\,\mathrm{d}t=\frac{\pi^4}{4}\,{}_4F_3\left[\begin{matrix}\frac12&\frac12&\frac12&\frac12\\&1&1&1\end{matrix};1\right]=4\pi^2L(f_{8.4.a.a},2),\end{aligned}\tag{43}$$

[3]For the second equalities in (43) relating Bessel moments to hypergeometric evaluations, see [BBBG08, (47)], sequence A263490 in [OEI26] and [BBBG08, (73)], respectively. The identifications between the latter and the critical values of the L-functions of the relevant modular forms can be found in [AMNT26, Zag18].

$$
\begin{aligned}
D_{4,\mathrm{odd}} &= \int_0^\infty I_0(t)K_0(t)^3\,\mathrm{d}t \stackrel{?}{=} 8L(f_{8.4.a.a},3),\\
D_{5,\mathrm{odd}} &= \int_0^\infty I_0(t)K_0(t)^4\,\mathrm{d}t \stackrel{?}{=} \pi^2 L(\chi\cdot\mathrm{Sym}^2\,g,2) = 2^3\sqrt{15}L(\chi\cdot\mathrm{Sym}^2\,g,3),\\
D'_{6,\mathrm{even}} &= \int_0^\infty K_0(t)^6\,\mathrm{d}t \stackrel{?}{=} 108\pi^2 L(f_{24.6.a.a},4),\\
D_{6,\mathrm{odd}} &= \int_0^\infty I_0(t)K_0(t)^5\,\mathrm{d}t \stackrel{?}{=} 144L(f_{24.6.a.a},5).
\end{aligned}
\tag{44}
$$

Here, the equalities with an "?" in (44) have been checked numerically.

When $k=1$, following from the fact that $L(\chi,n)\sim 1$ if $n=-2a$ and $L(\chi,n)\sim\pi^{2a+1}$ if $n=2a+1$, Deligne's conjecture holds in this case. (The L-function $L(\chi,s)$ is also known as the Dirichlet beta function $\beta(s)$.) On the other hand, by Theorem 1.2.2, the identities in (43), and the vanishing $L(f_{24.6.a.a},3)=0$ of the central value due to the negative sign of its functional equation, the conjecture also holds for $k=3$ and $(k,n)=(4,3),(6,4)$.

k	rank	critical integers n	c_n
1	1	$1-2a$ $2+2a, a\geq 0$	$D_{1,\mathrm{odd}}=1$ $\pi^{2a}D_{1,\mathrm{even}}\sim\pi^{2a+1}$
$4r-1$	$2r$	$2r$ $2r+1$	$\pi^{-r(r+1)}D_{k,\mathrm{even}}$ $\pi^{-r(r-1)}D_{k,\mathrm{odd}}$
$4r$	$2r$	$2r$ $2r+1$ $2r+2$	$\pi^{-r(r+1)}D_{k,\mathrm{odd}}$ $\pi^{-r(r+1)}D_{k,\mathrm{even}}$ $\pi^{-r(r-1)}D_{k,\mathrm{odd}}$
$4r+1$	$2r+1$	$2r+1$ $2r+2$	$\pi^{-r(r+1)}D_{k,\mathrm{odd}}$ $\pi^{-r(r+1)}D_{k,\mathrm{even}}$
$4r+2$	$2r$	$2r$ $2r+1$ $2r+2$ $2r+3$ $2r+4$	$\pi^{-r(r+3)}D_{k,\mathrm{odd}}$ $\pi^{-r(r+3)}D'_{k,\mathrm{even}}$ $\pi^{-r(r+1)}D_{k,\mathrm{odd}}$ $\pi^{-r(r+1)}D'_{k,\mathrm{even}}$ $\pi^{-r(r-1)}D_{k,\mathrm{odd}}$

TABLE 1. Critical integers n and values of c_n ($r\geq 1$)

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